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Master's Thesis

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Perceived Financial Well-Being and Mental Health

A compartmentalizing analysis

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Abstract

Policy makers and financial institutions face an issue with supporting financial well-being. This is in part complicated by its lack of accepted definition in the academic literature. It is well-established that perceived financial well-being is a major contributor to overall well-being, outweighing job satisfaction, relationship support satisfaction, and health status combined.

This study aims to increase specificity in the existing literature by dividing perceived financial well-being into its constituents and looking at their relationship to mental health, a subset of overall well-being. Specifically, we investigate how perceived present/future financial freedom/security impacts mental health. In this context, perceived financial well-being is defined as being able to meet present financial obligations, feeling secure in the financial future and being able to make choices that allow you to enjoy life.

We tested the hypothesis that higher levels of the constituents of perceived financial well-being lead to greater mental health. We performed exploratory factor analyses on CFPB survey data in order to define factors as constituents to perceived financial well-being along with mental health. We then used these factors to compare various structural equation models in order to assess the direction of relationships. Lastly, we used the factors to score the original dataset in order to estimate linear regression models, that allowed for the comparison of partial explanatory power of factors, along with evaluating the importance of the present and the future.

Our results indicate a significant, positive correlation between perceived financial well-being and mental health. Individuals that are able to meet obligations, feel more financially secure and free, tend to have improved mental health. The subset of future financial freedom, interpreted as financial confidence, constitutes the majority of explained variance in mental health. Perceived financial well-being explains 35% of the variance in mental health. Moreover, future financial well-being is deemed comparably more important than present financial well-being due to individual aversion to ambiguity and individual naivety towards own present-bias.

These results suggest that increasing financial confidence has the greatest effect on increasing mental health, compared to other constituents. Policy makers and financial institutions may find that these results assist in choosing the exact area of focus for increasing the well-being of consumers.

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This thesis is written in collaboration. The analysis was created following discussions in the group. In accordance with the rules, we specify the "main responsibility" for each chapter below.

Anders Mayland Larsen (rgn501): 1, 2.1.3, 2.1.4, 3.2, 3.3, 3.4, 4.2, 4.4, 4.5 (partial conclusion), 5.1 (partial conclusion), 5.2, 6

Mathias Amby Hansen (rdl367): 1, 2.1.1, 2.1.2, 2.2, 2.3, 3.1, 4.1, 4.3, 4.5 (partial conclusion), 5.1 (partial conclusion), 5.3, 6

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1 Introduction

Financial distress is shown to explain more of the variance in overall well-being than job satisfaction, relationship support satisfaction, and health status combined. This is the result of a recurring study conducted by the consulting firm PwC (2021) which concluded that 58% of its employees believe this to be true for themselves. Moreover, 63% of their employees feel their financial stress has increased since the start of the SARS-CoV-2 pandemic, alluding to the timeliness of this problem.

Various empirical studies have examined the relationship between perceived financial well-being and overall well-being (Netemeyer, Warmath, Fernandes, & Lynch, 2018), (Trombitas, 2012), (Barrett, Hogreve, & Brüggen, 2021), (Brenner, Meyll, Stolper, & Walter, 2020). These studies continually show how financial well-being is an important explanatory variable for overall well-being. Despite this, the importance of the underlying constituents of perceived financial well-being and its specific relationship to overall well-being is yet to be determined. In order to analyze said constituents, we rely on a large American data sample, collected by the Consumer Financial Protection Bureau (CFPB) and their resulting reports.

Although perceived financial well-being is increasing in the literary awareness, the focus on its exact correlation to certain specific components of overall well-being has received insufficient attention. The implications of prior studies on the topic deserve further attention in order to identify the importance of the constituents of perceived financial well-being and how they are associated with mental health. We analyze the effect from the constituents of perceived financial well-being on mental health. Specifically, we answer the research question:

How do the constituents of perceived financial well-being impact mental health?

In order to answer this question, we formally test the following hypotheses:

H₁: Perceived financial well-being correlates positively with mental health

H₂: Perceived financial well-being is a plausible, causal determinant of mental health

H₃: Perceived financial well-being regarding the future has a greater effect on mental health than that of the present

We first discuss several examples of survey-based research into the relationship between perceived financial well-being and overall well-being. We then present a behavioral

economic framework from which to view and model the problem, including findings from the behavioral finance literature. Furthermore, we establish the methodology of factor analysis and structural equation modeling which is used to structure, validate and compare empirical models. We utilize these methods to identify and define the constituents of financial well-being, mental health along with assessing their relationship and arguing for the likely causal path. Finally, we present our results, how they relate to existing research and what practical implications may be derived by policy makers, financial advisors and other financial agents.

1.1 Defining Perceived Financial Well-Being and Mental Health

Financial well-being as a measure is relatively new, and no single accepted, standardized definition exists. This lead to the proposition of developing a scientific definition which would benefit policy makers, academics, financial advisors etc. (Netemeyer et al., 2018).

In 2015, the Consumer Financial Protection Bureau (CFPB) published a report where a standardized approach to financial well-being was formulated. CFPB is an independent bureau that provides open access financial research, aiming to improve financial education for institutions and individuals (CFPB, 2021). One specific area of interest is research within perceived financial well-being. As of 2021, CFPB has published two reports on American financial well-being, which examine overall tendencies, and key point indicators resulting from individual financial well-being scores. Reports and data from CFPB serve as the foundation for our analysis. Based on qualitative research on American citizens, CFPB concluded that financial well-being can be divided into four dimensions (CFPB, 2015).

Figure (1.1) The four elements of Financial Well-Being

	Present	Future
Security	Control over your day-to-day, month-to-month finances	Capacity to absorb a financial shock
Freedom of choice	Financial freedom to make choices to enjoy life	On track to meet your financial goals

Source: CFPB (2015)

Present security (PS): Groups variables related to short-term financial expenses. This includes the ability to pay bills on time and not acquiring unreasonable debt.

Present freedom (PF): Groups variables that relate to short-term financial restraints, along with those that give the respondent an increased quality of life. This includes

vacations, continuing education, or re-balancing the work/leisure ratio.

Future security (FS): Groups variables related to unexpected financial expenses or exogenous shocks. FS measures the ability of respondents to withstand financial shocks by examining saving behavior or credit standing.

Future freedom (FF): Groups variables related to the respondents long-term financial goals. Financial goals could include lowering already existing debt within a reasonable time-frame, or saving towards retirement.

Combining these four dimensions, CFPB offers the following definition for perceived financial well-being (CFPB, 2015):

"We found that consumers perceived financial well-being as a state of being wherein a person can fully meet current and ongoing financial obligations, can feel secure in their financial future, and is able to make choices that allow them to enjoy life."

Furthermore, we wish to examine how these four dimensions affect the mental health of respondents, which entails a similar definition. Based on measurements listed by the World Health Organization (WHO), we use the following definition for Mental Health (WHO, 2018):

"Mental health is a state of well-being in which an individual realizes his or her own abilities, can cope with the normal stresses of life, can work productively and is able to make a contribution to his or her community"

To summarize, WHO, similarly to CFPB, defines and compartmentalizes mental health as consisting of four elements which are

- Realizing potential
- Stress
- Productivity
- Community contribution

We note that these elements are not easily measured. Measuring the difference in performed ability and potential ability is practically impossible. Contrarily, asking about perceived level of stress and measures of productivity are more easily attained. In addition, the definition leaves room for interpretation. The ability to contribute to your community is different from actually contributing. Moreover, what may seem a contribution to some, may be seen as detrimental to others. Nevertheless, we use this definition as the guideline for our further analysis.

1.2 Methodological procedure

We want to establish a modeling framework wherein the connection between the concept of perceived financial well-being and mental health can be measured. The modeling framework is illustrated in appendix A.1.

Firstly, we load raw survey data from CFPB, which is then cleaned to account for statistical inconveniences or to improve the data quality. Then, we specifically define the concepts of financial well-being and mental health, within our data, by utilizing the definitions stated by CFPB and WHO. Thereafter, we proceed to the factor generation process, where we first divide financial well-being into the four constituents shown in figure 1.1, by selecting variables that fit the description of each constituent. These are further decomposed into the final factors. To analyze if correlations exist between the factors, we perform a confirmatory factor analysis to ensure that further analysis is justifiable. With various defined factors encapsulating financial well-being and mental health, we score and enrich the original dataset, with additional variables for each factor. The next step is to examine which proposed causal path we deem to best fit the data. In this step, four structural equation models are proposed and compared, such that we continue the analysis with the best performing model. This model is then estimated by implementing ordinary least squares. We analyze the factor distributions across population variables to further validate the model. Lastly, alterations to the model are made to examine whether present-bias exists, and how subjective indicators improve the model quality.

1.3 Relationship to existing literature

The purpose of this paper is to clarify how perceived financial well-being, divided into present/future and freedom/security aspects, impacts mental health. Various recent studies have focused on perceived financial well-being and its correlation to overall or general well-being. However, the literature lacks an in-depth assessment of the constituents behind financial well-being and how these constituents compare with regards to importance. The majority of existing literature relies solely on objective indicators, such as income or other monetary measures when estimating overall well-being (Dolan, Peasgood, & White, 2008). As stated by Diener (1984), it is problematic to only account for objective measures when analyzing a measure that relies on deeply personal traits, such as cognitive judgment or life experience. Recently, subjective factors have gained increased attention with more studies including a mixture of objective and subjective measures. Furthermore, the continued use of proxies for overall well-being seemingly

creates a problem these studies try to fix: the lack of a standardized definition. CFPB provides a solid definition of perceived financial well-being, but seeing how it relates to another phenomenon with no clear definition seems counter-productive. For this reason, we utilize an established definition of mental health for our analysis.

Financial well-being, or lack thereof, is widely regarded as one of, if not the single greatest source of stress or lack of overall well-being in individuals. Several studies have examined which causes of stress are most prevalent for individuals, where stress associated with finances often ranked highly (Trombitas, 2012). Financial well-being partly consists of the propensity for savings. The importance of consumer saving motivation in the short and long run is largely derived from Maslow's (1943) hierarchy of need prepotency. Short-term, basic needs such as duly payment of bills relate to stress concerning immediate "deficiency needs". Conversely, long-term, complex needs such as retirement planning relate to "remote needs". Individuals face a trade-off between immediate utility and future utility. However, individuals often exhibit time-inconsistent, irrational behavior as not taking their future selves into account (Hershfield, Cohen, & Thompson, 2012).

Netemeyer et al. (2018) present a more compact definition of financial well-being than what is presented by CFPB. Their definition considers only the time aspect of financial well-being but not the distinction between financial freedom of choice and financial security as presented by CFPB (CFPB, 2015). Nevertheless, their findings indicate that 39% of the variance in overall well-being may be explained by financial well-being. The delineation made by Netemeyer et al. (2018) is traceable in other studies relating to financial well-being (Barrett et al., 2021) (Brenner et al., 2020). Barrett et al. (2021) analyze both financial well-being and mental well-being, but lack the link between these concepts, which we aim to establish. They examine how the pandemic affects these types of well-being, and find that resilient individuals have higher financial and mental well-being. The use of "overall well-being" as a measure is not well-defined and may be subject to much difference in interpretation. We counter this issue by applying an existing definition of mental health, created by the World Health Organization (WHO).

Further definitional inconsistencies occur when comparing the type of measurement used to conduct mental health analyses. As previously mentioned, Netemeyer et al. (2018), Barrett et al. (2021), Brenner et al. (2020), and CFPB (2017) are grounded within the same foundational framework, but with certain differences that alter the final definition of financial well-being. A supplementary measurement to mental health is

presented by Robb (2017), where subjective well-being is the primary driver in explaining mental health, motivated by the argument that it encapsulates individual factors related to life satisfaction (Diener, 2000). Robb (2017) found that financially stressed students generally exhibited lower subjective well-being, thus supporting the same hypothesis as with mental health.

Statistics Denmark (2018) has developed an indicator for the "economically vulnerable" based on five survey items reflecting perceived economic situation. They estimate that 8% of the adult population in Denmark are economically vulnerable, where young adults make up the majority. The elderly are generally not economically vulnerable. The data is gathered in collaboration with the EU-SILC operation (Statistics on Income and Living Conditions) (Eurostat, 2019). The survey questions show some resemblance to those used in a shorter, less reliable version of CFPB's survey, but notable differences exist and the means for reaching exactly those questions seem arbitrary compared to the CFPB approach. Nevertheless, the indicator is measurable and comparable across time. However, due to the limitations of the definition of financial well-being, the results are not comparable to those presented on the basis of CFPB's definition.

In a report on economic deterioration of young adults from 2018, the Danish Consumer Council (DCC) conclude that 11.9% of people between ages 18 and 30 experience repeated difficulty paying bills (The Danish Consumer Council, 2018). The report emphasizes the need to control for age within analyses of financial well-being. The data suggests that the financially struggling young adults exhibit significantly lower well-being and general health. The DCC poses the question whether this is a reaction to, or the prerequisite of, poor financial well-being, even though they made a similar analysis 3 years prior, making causal proposals from a time-difference perspective possible. However, the authors propose different policy changes to combat a problem which they fail to identify whether is a cause or a symptom. Identifying causal relationships is difficult. Nevertheless, tests may be applied as arguments to strengthen a causality hypothesis. We attempt to identify the causal relationship between financial well-being and mental health.

By answering our research question we contribute to the field, firstly by presenting and validating definitions, thus further corroborating a common conceptualization of financial well-being and mental health. The lack of an accepted definition is frequently mentioned in the literature. To counteract this potential weakness, we choose to clearly follow the guidelines and definitions created by CFPB (2017) and WHO (2018) when analyzing our data. Secondly, we contribute to the field by testing proposed causal links

that explain why and which constituents of financial well-being are major contributors in explaining mental well-being for the population.

We have not been able to find other studies that define mental health in accordance with WHO (2018), and as such, this study proposes a new angle to examine financial well-being and mental health. In essence, we specify the relationship between financial well-being and general well-being further. Most importantly, we identify the major underlying factors of perceived financial well-being.

2 Theoretical foundations

2.1 Findings from Behavioral Finance

Behavioral economics consists of positive theory, identifying where the normative theories of classical economics do not properly predict the behavior of economic agents. Concerning financial well-being, subjective measures have gained popularity (Robb, 2017) due to their greater explanatory power. In order to fully encapsulate and understand individual behavior, alternative theories proposed by behavioral economists prove useful. This is especially true when trying to understand the underlying tendencies and patterns within survey data consisting of individual perception. These concepts are therefore used to describe the framework we operate in, as well as provide possible explanations to findings later on. We implement these concepts within a single model, proposing a relationship between financial well-being across its time aspects, the translation from actual financial well-being to perceived financial well-being and mental health.

2.1.1 Prospect Theory

A majority of behavioral finance theory can be traced back to Kahneman and Tversky's (1979) analysis of behavior under different types of risk, later named Prospect Theory. Financial decisions of varying types include some degree of uncertainty and risk, indicating that understanding and applying prospect theory is beneficial when examining our problem. Kahneman and Tversky (1979) proposed a utility function with different properties from Expected Utility Theory (EUT). Their proposal was a value function and specific probability weighting, which aimed to explain the shortcomings of models assuming rational, risk-averse agents.

Kahneman and Tversky's experimental setup consisted of different lotteries, that had a closed-form solution under classical EUT. Consider the following two lotteries (Kahneman & Tversky, 1979):

A: (4,000 ; .80) or (3,000 ; 1)

B: (-1,000 ; .50) or (-500 ; 1)

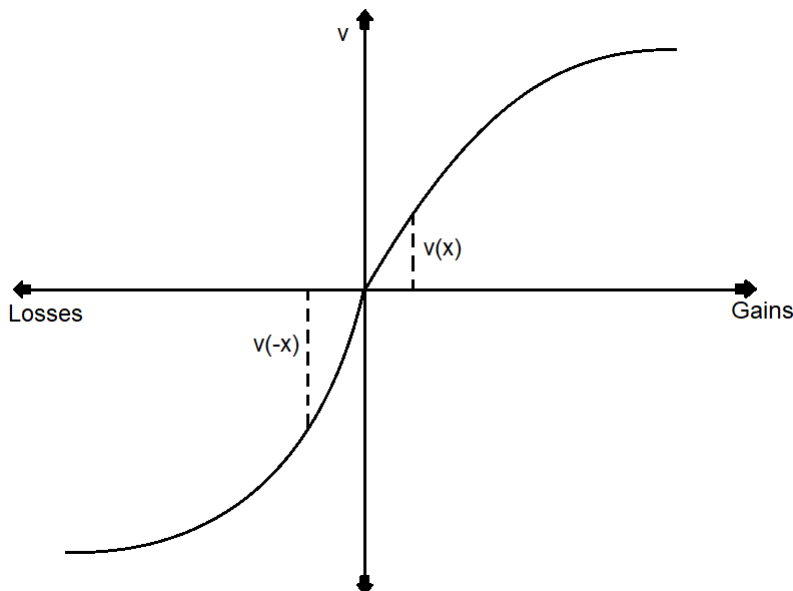
where participants in each lottery faced a choice between two options. In lottery A, participants could either choose 4,000 units with probability $P = .80$, and 0 with $P = .20$, or the other option which included 3,000 units with probability $P = 1$. Lottery B introduced a loss, where participants could choose to lose 1,000 units with probability $P = .50$ and break even with $P = .50$ or face a guaranteed loss of 500 units with probability $P = 1$. Actual empirical evidence suggests that respondents

do not act according to EUT, since the majority (84%) chose the risk-free option in lottery A, and (69%) chose the risky option in B (Kahneman & Tversky, 1979). Under EUT, respondents should choose the risky option in A ($U(3, 200) > U(3, 000)$), and be indifferent in B ($U(500) = U(500)$), since the expected unit yield is identical.

These inconsistencies with EUT lead Kahneman and Tversky to propose said value function $v(x)$, which has different properties, dependent on whether the respondent was facing a potential gain or loss. The properties of the value function are as follows:

- (i) $v(x)$ is continuous, strictly increasing and reference-dependent, i.e. $v(0) = 0$
- (ii) $v(x)$ is concave, whereas $v(-x)$ is convex, indicating risk-averse behavior in gain-domain, and risk-seeking behavior in loss-domain
- (iii) Individuals are loss-averse, i.e. $-v(-x) > v(x)$ for $x > 0$

Figure (2.1) Value function



Within this methodology, we establish a modeling framework and way of thinking, wherein deviations to classical theory can be explained and compared. The intuitive implications of prospect theory can be further extended to the financial well-being and mental health framework. Since financial decisions regarding the present, and especially the future has a high degree of uncertainty and risk, the foundations of prospect theory can provide insights into individual behavior.

Reference-dependence is important for any decision making process, where alternative scenarios are compared. Within financial well-being this may be reflected in future financial well-being by examining individual saving behavior. Should one put savings into a bank or credit union savings account, a retirement account such as a 401k or IRA¹, or invest the funds directly in securities? These alternatives are com-

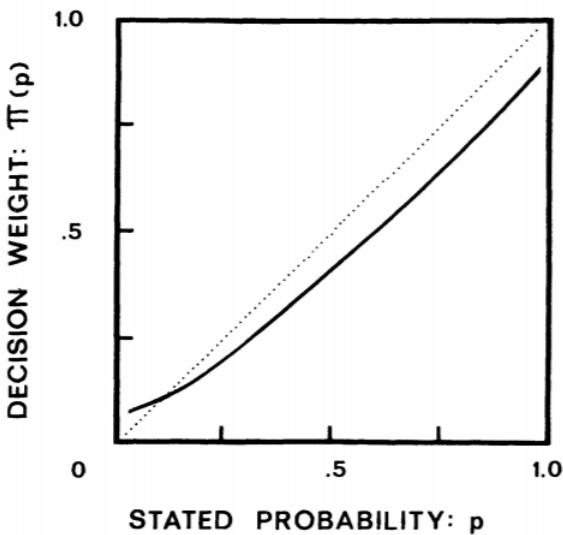
¹401K and IRA (Individual Retirement Account) are the most common forms of tax-advantaged

pared to the status-quo, which is to hold the funds. The concave properties for $v(x)$ in the gain-domain indicates that people tend to prefer safer options, rather than risky, even though the risky option may have the potential to yield a higher value.

Financially stable individuals are more likely to be in the gain-domain, and make safer financial decisions, further ensuring and securing their long-term financial stability. Contrarily, the risk-seeking behavior in the loss-domain, could contribute to explaining why financially vulnerable individuals tend to accumulate more debt, and worsen their financial well-being. Lastly, losses hurt more than equal gains benefit the individual. A given monetary loss is more detrimental for a financially vulnerable individual than the same monetary gain. This may worsen present financial security, since fixed expenses cannot be paid. Since financial stress has a large impact on mental health, the value function and individual risk-behavior can aid in understanding how and why mental health is affected.

In addition to the value function, Kahneman and Tversky found inconsistencies in the way individuals weighted specific probabilities. Their proposed probability weighting function $\pi(P)$ creates a connection between assigned decision weights and probabilities. Specifically, individuals tend to overweight small probabilities $\pi(P) > P$ and the degree of relative overweight increases for smaller probabilities. For smaller probabilities, $\pi(P)$ is defined as sub additive, where $\pi(cP) > c\pi(P)$, for $0 < c < 1$. The weighting function $\pi(P)$ is shown in figure 2.2 and illustrates how a decision weight is assigned to each probability P , and further how P affects the value of the given prospect.

Figure (2.2) Weighting Function



Source: Kahneman and Tversky (1979)

Assigning a relatively high weight to unlikely events can be witnessed and expressed retirement savings in the United States. 401K's are offered by employers to employees.

by the future security dimension. This is most notable with insurance, where individuals insure themselves towards a multitude of unlikely events. Thus, individuals protect themselves against *unexpected emergencies*, which have a small probability of happening.

2.1.2 Intertemporal decision-making

Risk-averse behavior in the gain domain is exemplified in Lottery A, where the lower yielding, safe option was preferred to the rational option. This type of behavior was introduced by Kahneman and Tversky and later observed across different studies (Halevy, 2008) (Mather et al., 2012), and is labeled the *certainty effect*. This phenomenon can be applied to intertemporal decision making, which we examine in the analysis. The distinction between present and future dimensions can be interpreted in terms of certainty, where the present is considered relatively certain, and the future is considered relatively uncertain. Arguably, the safe immediate gratification is preferable, other things equal, and uncertain future events are undesirable, and should be discounted accordingly. Furthermore, because we operate with time-dimensional factors, a regular time-discounting component should be included. The following model extends the simple traditional discounted utility theory, where $\sum_{t=0}^{\infty} \delta^t u(x_t)$ is the maximized object (O'Donoghue & Rabin, 1997).

$$U = u(x_t) + \beta \delta u(x_{t+1}) + \beta \delta^2 u(x_{t+2}) + \beta \delta^3 u(x_{t+3}) + \dots \Leftrightarrow$$

$$U = u(x_t) + \beta \sum_{\tau=1}^{\infty} \delta^{\tau} u(x_{t+\tau}),$$

where it is evident, that the standard model with exponential discounting is captured as a special case where $\beta = 1$. However, given that we only view time in a two-dimensional framework (present or future), the discounting parameter, δ^{τ} , is not exponential. In addition, we wish to focus on mental health, therefore we rewrite the equation to account for this:

$$MH_i = mh(x_{i,present}) + \beta mh(x_{i,future}) \quad (1)$$

where MH_i measures the total mental health for respondent i , $mh(x_{i,j})$ measures respondent i 's mental health for a given time-dimension j , $\delta = 1$ is assumed, and β can be interpreted as present-bias or certainty effect. In utility models, β is in $[0, 1]$ indicating that future utility cannot be greater than present (O'Donoghue & Rabin, 1997). However, we want to allow the future to have a greater impact on mental health than the present, which we test.

We note that we are dealing with *perceived* mental health, and as such potential discrepancy between actual and perceived mental health may be present. Kahneman, Wakker, and Sarin (1997) proposed an experienced-utility idea which was later simplified and modeled by Gardner and Oswald (2007), proposing a framework to account for the potential discrepancy due to individual perception which we apply in the following way:

$$MH_i = h(f(PF_i, PS_i, FF_i, FS_i)) + \varepsilon_i \quad (2)$$

where MH_i is a measure of perceived mental health, f is the true financial well-being of an individual as a function of financial present freedom, present security, future freedom and future security, h is a non-differentiable function translating actual financial well-being to perceived financial well-being and ε_i is the error consisting partly of omitted variables that impact mental health such as general health, marital status etc. This describes the way in which we conceptualize mental health and its relation to perceived financial well-being. The function f might be extended to further clarify the presence of the present-bias parameter β . We note that h is an unobserved function and that the translation from perception to reality is not a topic of this paper. However, it should be noted that a potential issue of examining data consisting of perceptions and opinions may lead to respondents not perfectly reflecting the truth.

2.1.3 Ambiguity aversion

A natural continuation to the prior example of modeling the expected present bias, and arguing for a case where agents are either naive or sophisticated, is to answer the question, *why* the present bias exists. A simple explanation is the proclivity of individuals to discount future payoffs. This explanation is alluring given its simplicity as well as its repeated use in economic modeling and ease of applicability (Kreiner, Leth-Petersen, & Willerslev-Olsen, 2020). However, discounting future payoffs requires complete information of these payoffs and their respective probabilities. Individuals do not always have this information with regards to their financial well-being. It is safe to assume that individuals have near complete information of their present financial well-being. Contrarily, future financial well-being is a function of many factors exhibiting uncertainty as we will later see. What is the exact probability of incurring unexpected expenses or emergencies? What is the exact probability of meeting the financial goals you are planning for in the present? The future is ambiguous in many ways and individuals exhibit peculiar properties when facing ambiguity, which EUT fails to model due to unknown probabilities.

Given this aversion towards ambiguity, we assume that perceived future financial well-being has a greater impact on mental health than that of the present. Gilboa and Schmeidler (1989) presented a possible framework for modeling ambiguity aversion, based on the following idea:

"One conceivable explanation of this phenomenon which we adopt here is as follows: ...the subject has too little information to form a prior. Hence (s)he considers a set of priors as possible. Being [ambiguity] averse, (s)he takes into account the minimal expected utility (over all priors in the set) while evaluating a bet."

In essence, the assumption is that subjects facing ambiguity expect the worst and try to make the best of that situation. Their model is fittingly named Maxmin Expected Utility (MEU). Given the assumption that individuals have complete information of their present financial well-being, their perception should reflect reality. This reality may be positive or negative. However, since the future is ambiguous and individuals tend to expect worst case scenarios, individuals are biased negatively with respect to their financial future. We expect this to partly explain why we assume perceived future financial well-being to have a greater impact on mental health than that of the present. Considering the relation of ambiguity aversion to the above intertemporal decision making model with quasi-hyperbolic discounting, the β of the model, symbolizing present-bias may easily be interpreted as the level of ambiguity aversion.

2.1.4 Naive versus sophisticated agents

Laibson (1997) showed how the above model of intertemporal decision-making has empirical implications. For $\beta < 1$, agents behave time inconsistently. The question then becomes: do agents realize their propensity for time inconsistent behavior and aversion to ambiguity? Do they take into account that the well-being of the future becomes the well-being of the present from the perspective of the future?

We assume that agents belong to one of two categories. *Naive* agents are present-biased and unaware of it. They wrongfully believe that $\hat{\beta} = 1$ for future periods. An example of a naive agent might be a thesis student who wants to work on his thesis this week, because his well-being is better for it in the future. However, he is present-biased and tends towards immediate well-being. In return, he pushes the work to next week, thinking that he will write his thesis then (belief that $\hat{\beta} = 1$ next week). However, he is exactly as present-biased next week and continues to postpone his writing, resulting in low levels of well-being if he reaches the last week before the hand-in deadline.

Sophisticated agents are just as present-biased as the *naive* agents. However, they take into account that their future selves are just as present-biased. Therefore, they correctly predict their future behavior if they postpone writing their thesis this week. In relation to our problem, we hypothesize that a portion of agents are at least partially naive and therefore fail to realize the importance of future financial well-being on their mental health due to being present-biased or ambiguity averse. The prior example of present-bias in thesis writing may easily be reformulated to describe the behavior of naive agents saving up for retirement.

2.2 Factor analysis

In order to define the previously mentioned unobserved dimensions, we utilize factor analysis which is a multiple regression analysis of the following form (Andersen & Milhøj, 2020):

$$\mathbf{X} = \mathbf{B}\mathbf{F} + \mathbf{E}, \quad (3)$$

where $\mathbf{X}$ is an $n \times 1$ column vector of stochastic variables $X_1, \dots, X_n$, $\mathbf{B}$ is an $n \times k$ matrix of factorloadings, $\mathbf{F}$ is a $k \times 1$ column vector of unobserved stochastic factors, called the factor scores and $\mathbf{E}$ is a column vector of independent error terms called the unique component.

Our framework entails that $\mathbf{X}$ measures the mental health of respondents, based on predefined factors $\mathbf{F}$ with factorloadings $\mathbf{B}$ and unobserved errors. Because neither mental health or the financial well-being factors are directly observable, we end up estimating a model consisting of 5 unobserved (latent) variables. Each latent variable in the model depends on prerequisite manifest variables which are directly observable for each respondent. Each of the N respondents answer n questions representing manifest variables, which are connected to a specific latent factor.

$\mathbf{B}$ and $\mathbf{F}$ are then estimated based on the empirical covariance matrix.

$$\mathbf{S} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T,$$

which is a sufficient sample function for the theoretical covariance matrix.

$$\mathbf{V}_X = \mathbf{B}\mathbf{V}_F\mathbf{B}^T + \mathbf{V}_E = \mathbf{V}_X(\theta_1, \dots, \theta_p), \quad (4)$$

This function is solved by maximum likelihood estimation, which gradually improves

the parameter values by using iterative measures, until the maximum is found. On rare occasions the communality restriction cannot be fulfilled, meaning that the maximum likelihood estimate is beyond the parameter space. In such scenarios, a corner solution called *Heywood* is used, which enables the analysis to still be viable.

After estimation, the initial factor patterns are then altered by utilizing factor rotation. Consider the extension to the product $\mathbf{BF}$:

$$\mathbf{BF} = \mathbf{BAA}^{-1}\mathbf{F},$$

where $\mathbf{A}$ is a $k \times k$ matrix, which indicates that $\mathbf{A}^{-1} = \mathbf{A}^T$. This result means that any random matrix $\mathbf{A}$ can be inserted, which is utilized when we perform an oblique rotation of the factor patterns with **PROMAX**. This type of rotation allows for factor correlation, which will become essential for the analysis later on. The alternate rotation method orthogonal does not allow factor correlation to be present, and is therefore harder to implement in practice. Rotations of the coordinate system are made to better illustrate and interpret each factor, such that variables with a high correlation are grouped closely together to an axis indicating the related factor. Factor rotation does also improve the replicating capabilities if the matrix $\mathbf{A}$ is known. We choose to focus on oblique rotations in the analysis, because we want to allow for correlations between factors.

2.2.1 Second-order factor analysis

When formulating a model that includes a multitude of latent factors which correlate with each other, one could introduce the concept of a second-order factor analysis. This analysis is implemented by conducting a first-order factor analysis, as described above, on the calculated factors, thus constructing more general factors than before (Jöreskog, 1970). The intuition and interpretation of the second-order factors are more generalized and abstract. In the following, we group time-dependent factors together, to construct *present* and *future* factors, which each consists of four specific factors, to examine which time dimension that has the largest impact in explaining respondents mental health.

As mentioned before, the oblique rotation-method **PROMAX** allows for factor correlation, which is crucial for the second-order analysis. Were another orthogonal rotation method used, i.e **VARIMAX**, we would estimate the already-existing factors, yielding an identical result. When factor correlation is present we instead estimate different factors based on said correlations. This procedure can be applied to higher dimensions than two, continuously creating more general factors (Jöreskog, 1970).

2.3 Structural equation modeling

A supplementary method to the factor analysis is to construct a structural equation model, where the purpose is to formulate a model that explains the sequence of the correlations. This model thus serves as an alternate way to intuitively think about a possible causal correlation. Constructing models with manifest variables is distinctly different to those with latent variables. Because we generate unobserved latent factors, we focus on the theoretical literature related to models with latent variables.

One must specify the linear equation system including definitions for each latent variable. Dependent on the type of model, some of these equations may include both manifest and other latent variables. One must specify equations for both standard deviations and covariances. The models parameterization is performed by standardizing the manifest variables. Consequently, the last coefficient within each latent factor grouping equals 1. Consider the following model, where mental health is predetermined by 8 factors:

Linear Equations

$$\begin{aligned} f_{MH} = & pf_{MH,JoF} \cdot f_{JoF} + pf_{MH,Mat} \cdot f_{Mat} + pf_{MH,FCONT} \cdot f_{FCONT} + \\ & pf_{MH,Bud} \cdot f_{Bud} + pf_{MH,FPL} \cdot f_{FPL} + pf_{MH,FCONF} \cdot f_{FCONF} + \\ & pf_{MH,UEX} \cdot f_{UEX} + pf_{MH,UEM} \cdot f_{UEM} + d_1 \end{aligned}$$

std

$$\begin{aligned} f_{JoF}, f_{Mat}, \dots, f_{UEM} &= \mathbf{V}_{JoF}, \mathbf{V}_{Mat}, \dots, \mathbf{V}_{UEM} \\ d_1 &= \mathbf{V}_{d1} \end{aligned}$$

covariance

$$\begin{aligned} f_{MH} \cdot f_{Mat} &= \mathbf{S}_{MH,Mat} \\ f_{MH} \cdot f_{FCONT} &= \mathbf{S}_{MH,FCONT} \\ \dots &= \dots \\ f_{UEX} \cdot f_{UEM} &= \mathbf{S}_{UEX,UEM} \end{aligned}$$

where f_i denotes factor i, $pf_{i,j}$ denotes the parameter estimate between factor i and j, d_1 denotes the residual, $\mathbf{V}_i$ denotes the variance related to item i, and $\mathbf{S}_{i,j}$ denotes the

covariance between factor i and j .

The model above proposes a simple correlation path, where mental health is explained in a single step by the remaining factors. For simplicity we restrained from defining each factor and covariance in detail, since it would add 36 equations to model above². The simple model above states that a single path should explain the correlation, but more complex models could add additional steps, thus increasing the number of equations in the model. Furthermore, model modifications are easily observable and applicable. If any correlations between latent variables are not statistically significant, one could nullify the specific correlation and execute the modified model. This type of model thus provides a way in which a step-wise optimization can be achieved.

When measuring and comparing different SEM models, certain fit statistics such as the ratio between χ^2 -score and degrees of freedom, Root Mean Square Error of Approximation (RMSEA), Adjusted Goodness of Fit Index (AGFI) and Comparative-Fit index (CFI) are applied. RMSEA and CFI are both tied to the fit function $\tilde{F}$, which is defined as the following.

$$\tilde{F} = \log \left| \sum(\omega) \right| + tr \left(\mathbf{S} \left(\sum \right)^{-1} (\omega) \right) - \log |\mathbf{S}| - p \quad (5)$$

where $\mathbf{S}$ is an unstructured covariance matrix, $\sum(\omega)$ is a model-implied covariance matrix, and p is the number of variables (Xia & Yang, 2019). From equation (5) we note that the fit function $\tilde{F} \geq 0$ when the empirical covariance matrix is equal to the fit matrix. The χ^2 - DF ratio enables a quick and easy comparison between the significance of all models. With structural equation models a lower estimate is preferable, because a significant model indicates rejection. RMSEA is an error measure, indicating how far a hypothesized model is from the "perfect" model, meaning that values closer to 0 are preferable. The error is calculated as the ratio between the hypothesized Fit-function and corresponding degrees of freedom.

$$RMSEA = \sqrt{\frac{a_H \cdot \tilde{F}_H}{df_H}}$$

where a_H is a scaling parameter which allows for inclusion of sample size adjusted chi-square values. Generally RMSEA values under 0.08 are valued as a good fit for the given model. AGFI compares the variance from the formulated model and the actual covariance matrix, and indicates the proportion in between. Ideally, the hypothesized model should encapsulate all actual variance, meaning that a higher AGFI estimate is

²For the full model description, see Appendix B

desirable. A cut-off point for a good fit is an estimate around 0.9, i.e 90% of the actual variance is encapsulated by the model.

CFI compares the fit between the hypothesized model and the actual data. CFI is preferable to other fit measures because it is not sample size sensitive, due to the inclusion of scaling parameters. CFI values above 0.9 is generally accepted as a good fit (Hooper, Coughlan, & Mullen, 2007).

$$CFI = 1 - \frac{a_H \tilde{F}_H}{a_B \tilde{F}_B}$$

where a_B and $\tilde{F}_B$ is a shifting parameter and fit-function for the baseline model, respectively (Xia & Yang, 2019).

3 Data and Factor generation

3.1 Data overview

The latest report analyzing financial well-being is based on survey-data gathered by CFPB (CFPB, 2017). This dataset is openly available to the public, and serves as an extension to the previous report from 2015 (CFPB, 2015). As CFPB gradually releases new publications relating to financial well-being, the data is updated as well. This analysis reflects data for a specific period in time, and conclusions relating to causality or estimates may differ for future installments.

The survey-data is constructed as cross-sectional data, where each observation corresponds to individual responses measured across 217 variables. Decomposing the variable list, 175 variables relates to actual survey-questions that respondents must answer, where the remaining 42 variables relates to indexes, summary scales, panel data, and personal identification variables. Any identification statistics in the data are merely artificial, and can therefore not be linked to a specific individual, therefore maintaining respondent anonymity. 10 of these variables are referred to as "CFPB survey questions". These are the culmination of CFPB's analysis, where they deem these 10 questions in conjunction to be the best representation of perceived financial well-being. The sample contains a total of 6,394 observations. We examine whether data has any demographic biases by examining the distribution for certain demographic factors.

It is evident from table 3.1, that the gender distribution is borderline uniform, with a slight overrepresentation of males (53%). Likewise, the age distribution, measured by *generation* is relatively evenly disbursed, with a large share of respondents in each generation. However, a slight overrepresentaion of *Boomers* (36%), and underrepresentation of *Pre-Boomers* (18%) is present. Likewise, in 2017 the US population had a high percentage of *Boomers* which, other things equal, increased the probability for a respondent to be within that generation (Statista, 2021b).

For the most part, *Education* within the sample size is fairly evenly distributed, where most respondents achieved a college degree (30%). As expected, the proportion of respondents who have longer graduate degrees (19%) and especially less than a high school degree (6%) is relatively small. The distribution of *Race / Ethnicity* has the largest skewness of all demographic factors. The distribution is clearly left-skewed towards, indicating a vast majority of White, non-hispanic respondents (73%). Even-though race is skewed, it is generally in accordance with the race distribution for the

Table (3.1) Demographic inspection

Variable	Observations	Percent
Generation		
<i>Pre-Boomer</i>	987	17.9
<i>Boomer</i>	1,970	35.8
<i>Gen X</i>	1,241	22.5
<i>Millenial</i>	1,313	23.8
Gender		
<i>Male</i>	2,943	53.4
<i>Female</i>	2,568	46.6
Education		
<i>Less than high school</i>	306	5.6
<i>High school degree/GED</i>	1,309	23.8
<i>Some college/Associate</i>	1,656	30.1
<i>Bachelor's degree</i>	1,217	22.1
<i>Graduate/professional degree</i>	1,023	18.6
Race / Ethnicity		
<i>White, Non-hispanic</i>	4,034	73.2
<i>Black. Non-hispanic</i>	508	9.2
<i>Other, Non-hispanic</i>	289	5.2
<i>Hispanic</i>	680	12.3
Marital Status		
<i>Married</i>	3,398	61.7
<i>Widowed</i>	309	5.6
<i>Divorced/separated</i>	573	10.4
<i>Never married</i>	915	16.6
<i>Living with partner</i>	316	5.7
Household Income		
<i>Less than \$20,000</i>	486	8.8
<i>\$20,000 to \$29,999</i>	418	7.6
<i>\$30,000 to \$39,999</i>	519	9.4
<i>\$40,000 to \$49,999</i>	393	7.1
<i>\$50,000 to \$59,999</i>	445	8.1
<i>\$60,000 to \$74,999</i>	571	10.3
<i>\$75,000 to \$99,999</i>	855	15.5
<i>\$100,000 to \$149,999</i>	1,019	18.5
<i>\$150,000 or more</i>	805	14.6

US population in 2017 (Statista, 2021a). Respondent marital status does not follow a clear interpretable scale, but the vast majority (62%) of respondents are married. Once again, these demographic tendencies are consistent with general population estimates (US Census Bureau, 2021). Household Income is also relatively evenly disbursed, with a slight overrepresentation of respondents in the middle/upper class. When adjusting for variable household size, we find that these results are in accordance with the general population (Pew Research Center, 2018).

Examining demographic factors and appurtenant descriptive statistics, any potential selection bias in the sample is not apparent. The data provided by CFPB is therefore viewed as representable for the US population, and suitable for further analysis.

3.1.1 Data cleaning

A hurdle in need of correction is respondents that have not answered questions we intend to use in the factor analysis. Values corresponding to "not answered" are scattered across several categorical values. The response labels and values range from *"Response not written to database"* (-4), *Refused* (-1) and *I don't know* (8). Because data is categorical and the values for not answering the question is not assigned to a single value, we choose to exclude observations wherein one of these options are present. Removing observations reduce the sample size from 6,394 to 5,210.

We later find that some variables do not correlate significantly with our factors. Therefore, we choose to reintroduce observations, where only the unused variables are not answered, in order to maximize the number of usable observations. This procedure increases the sample size from 5,210 to 5,511 observations.

3.2 Constituent variable selection

This section presents the methodology behind the variable selection for use in the further analysis. Furthermore, we describe the variables selected and how they attribute to the given constituent factor. Based on the dimension definitions, we proceed to further segregate each variable. The segregation is performed by manually examining each variable and appurtenant questions in the variable list (CFPB, 2017). If the variable fits within the definition of a specific dimension (e.g. present freedom), we proceed to include it as a manifest variable for the factor analysis. The complete listing of variables for each factor is described in Appendix A.

In order to estimate the correlation between perceived financial well-being and mental

health, we identify the variables that best fit the definition of each constituent. This process consists of a subjective iteration of the complete variable list with the purpose of attributing each constituent with relevant variables. Table B.1 in appendix A visualizes all constituents and the variables attributed to them.

The *present security* constituent is assigned to variables concerning control over short-term finances. These variables include, but are not limited to

- Budgeting
- Duly bill payment
- Level of self control
- Feelings of financial concern

We assume these variables in conjunction represent a factor concerning PS. We view budgeting as a means to increase awareness of finances and clarifies the need for correction. In essence, this increases the control over finances. Paying bills on time is a further sign of having control over your finances. Furthermore, greater self control possibly increases control over finances by decreasing the risk of impulsive spending and budget deviation. Lastly, some of the actual financial well-being survey questions fit the description of this constituent well and are included. We select these variables for factor analysis in order to test the hypothesis that these variables may correlate with a single, or perhaps multiple factors that represent facets of short-term financial control.

The *present freedom* constituent includes variables concerning ability to enjoy life despite limited financial resources. These variables include, but are not limited to

- Materialism
- Frugality
- Feelings of being controlled by one's finances

We assume these variables in conjunction represent a factor concerning PF. Materialism may relate to greater financial stress associated with increased spending on luxury items. Respondents who are more materialistic may feel more hindered by their limited finances compared to those that are more frugal.

The *future security* constituent includes variables concerning ability to absorb financial shocks. These variables include, but are not limited to

- Save habits
- Insurance
- Investing
- Feeling secure in one's financial future

We immediately see how factors are correlated. Respondents who do not have money left over at the end of the month (low PS) and do not feel able to enjoy life due to financial limitations (low PF) are less likely to have a savings account, invest in stocks and pay for insurance. This serves as evidence to suggested causal paths: present financial well-being determines future financial well-being.

The *future freedom* constituent includes variables concerning propensity to meet financial goals. such as having goals and a plan to follow said goals, having a pension or retirement account. These variables include, but are not limited to

- Goal planning
- Pension/retirement account
- Ability to work towards long-term goals
- Feelings of being behind with one's finances

The importance of balancing financial well-being in multiple domains is apparent. Respondents who have financial goals, work towards them and pay towards a retirement account are not at all financially robust if they do not pay attention to their future security and are unable to absorb a financial shock. The constituents of financial well-being work in conjunction. In example, respondents who display high levels of three constituents but pay no attention to one of them may be worse off compared to respondents who display lower overall levels but encompass all four domains.

3.3 Fitting factors to constituents

In this section, we present the variables for each financial well-being constituent factor as well as for the *mental health* factor. Furthermore, we conduct factor analyses for each constituent in order to identify which variables are important components for each. The factor analysis serves as the foundation for later models.

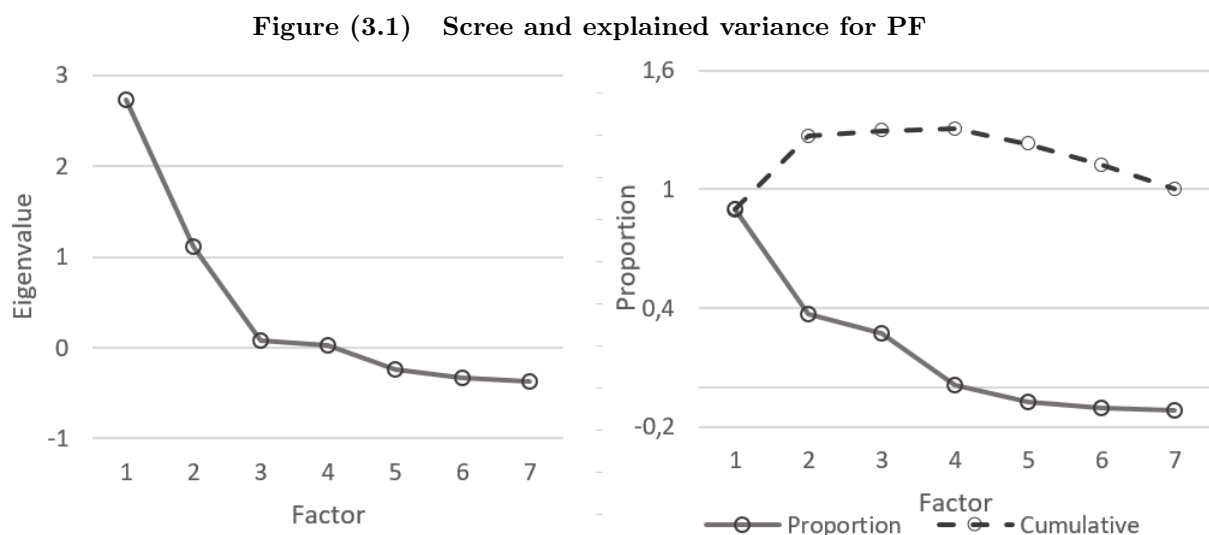
For each hypothesized dimension within financial well-being, we start by examining correspondent eigenvalues to ensure that the correct number of factors are included in the model. This means that we allow for a single constituent to consist of multiple underlying factors. However, we make the restriction that these underlying factors must correlate, otherwise it is difficult to argue that they jointly make up a single constituent. We inspect each variable within the given dimension and interpret the magnitude and sign for each correlation. If more than one factor is calculated for a given dimension, we further examine how these factors correlate with each-other.

Contrary to other literature utilizing factor analysis, we choose not to calculate and rely on Cronbahcs alpha when assessing the validity of each factor grouping (Barrett

et al., 2021). The way in which the questions and appurtenant categories are defined, limits the usefulness, because they are not formulated according to a Likert scale. We would thus need to restructure the categories for each variable in order for Cronbachs alpha to be a viable measure.

3.3.1 Identifying common factors

We begin by identifying common factors for the variables we believe match the constituent, *present financial freedom of choice* (PF). We apply factor-fitting methods as described in section 2.2. We find two eigenvalues greater than 1, meaning that two factors are fitted. We assess whether 2 factors seem sufficient.



It is evident from figure 3.1 that including additional factors will not explain significantly more of the variance for this constituent. We establish the null hypothesis that 2 factors sufficiently describe the variance in the chosen 7 variables against the null that more factors are needed. The χ^2 -test is of size $Q = 177$ with 8 degrees of freedom, resulting in a clear rejection of the null on a 5%-level of significance. This is no surprise since it seems practically unreasonable to find variables that encapsulates all relevant variance within *present freedom*. However, it is again evident from the scree plot that introducing more factors will not explain significantly more of the variance. As such, this test for model significance will not be repeated going forward, since the objective here is not to create a well-fitting model, but to identify significant correlations within the data. We look at the factor pattern in table 3.2.

From the factor pattern table we see how factor 1 significantly correlates with the three survey questions we deem to concern present freedom of choice. The signs of the correlations are in accordance with our expectations, where lower values for *FWB1_4*, and higher values for *FWB2_1* and *FWB2_4* indicate better *present financial freedom of choice*. This partial factor of PF is therefore interpreted as *joy of freedom (JoF)*.

Table (3.2) Factor pattern for PF

Variable	Label	Factor 1	Factor 2
BENEFITS_5	Work/Life Benefits	.	.
FRUGALITY	If I can re-use an item I already have, there's no sense in buying something new	.	.
MATERIALISM_2	The things I own say a lot about how well I'm doing in life	.	54*
MATERIALISM_3	I like to own things that impress people	.	93*
FWB1_4	I can enjoy life because of the way I'm managing my money	-67*	.
FWB2_1	Giving a gift...would put a strain on my finances for the month	85*	.
FWB2_4	My finances control my life	71*	.

Printed values are multiplied by 100 and rounded to the nearest integer. Values greater than 0.5 are flagged with an "*". Values less than 0.35 are not printed.

Factor 2 significantly, positively correlates with the two variables concerning what we interpret as overall *materialism* (*Mat*).

The two factors are positively correlated (0.11). Higher degrees of *materialism* thus correlates negatively (because of the reverse coding of the questions) with *joy of freedom*. If you like to impress people with and feel judged by your material possessions, you are likely more financially restricted since you may exhibit greater willingness to pay than those with low degrees of materialism. We note that work/life benefits and frugality are not fitted to a factor even though one might presume that frugality correlates negatively with materialism but the evidence for this is not strong enough in our data.

We continue by identifying common factors for the variables we believe match the constituent, *present financial security* (*PS*). We find 2 eigenvalues greater than 1, meaning that two factors are fitted and more factors do not contribute significantly to explaining variance. We look at the factor pattern in table 3.3.

Table (3.3) Factor pattern for PS

Variable	Label	Factor 1	Factor 2
COLLECT	Contacted by debt collector in past 12 months	49	.
EARNERS	Household members currently contributing to paying household bills	.	.
ENDSMEET	Difficulty of covering monthly expenses and bills	79*	.
FWB1_5	I am just getting by financially	59*	.
FWB1_6	I am concerned that the money I have or will save won't last	60*	.
FWB2_2	I have money left over at the end of the month	-76*	.
MANAGE1_1	Paid all your bills on time	-53*	.
MANAGE1_2	Stayed within your budget or spending plan	-44	44
MANAGE1_3	Paid off credit card balance in full each month	-62*	.
PROPPLAN_1	I consult my budget to see how much money I have left	.	75*
PROPPLAN_2	I actively consider the steps I need to take to stick to my budget	.	84*
SELFCONTROL_1	I often act without thinking through all the alternatives	.	.
SELFCONTROL_2	I am good at resisting temptation	.	.
SNAP	Any household member received SNAP benefits	40	.
VOLATILITY	Household income volatility	.	.

Printed values are multiplied by 100 and rounded to the nearest integer. Values greater than 0.5 are flagged with an "*". Values less than 0.35 are not printed.

From the factor pattern table we see how factor 1 significantly correlates with the three survey questions concerning PS along with debt collector contact, money-management

proficiency, ability to make ends meet, and SNAP³ benefit receiving. The signs of the correlations are in accordance with our expectations, such that having been contacted by a debt collector correlates positively with having difficulties paying monthly bills but correlates negatively with paying off credit card balance and having money left over at the end of the month. This partial factor of PS is therefore interpreted as *financial control (FCONT)*. Factor 2 significantly and positively correlates with the two variables concerning tendency to follow a budget. We interpret this factor as *Budgeting (Bud)*.

The two factors are negatively correlated (-0.22). A greater tendency to do *budgeting* correlates with lower levels of *financial control* (due to the reverse coding of some questions). This introduces the possibility that budgeting may help increase the control over your finances and a causal relationship is more likely since the questions concerning budgeting is about the present and some of the survey questions concern whether respondents believe money will last for the month (near future). We note that number of household earners, household income volatility and level of self control are not fitted to a factor.

We continue by identifying common factors for the variables we believe match the constituent, *future financial freedom (FF)*. We find 2 eigenvalues greater than 1, meaning that two factors are fitted and more factors do not contribute significantly to explaining variance. We look at the factor pattern in table 3.4.

Table (3.4) Factor pattern for FF

Variable	Label	Factor 1	Factor 2
FINGOALS	Do you have a current or recent financial goal?	47	.
FWB1_3	Because of my money situation...I will never have the things I want in life	.	72*
FWB2_3	I am behind with my finances	.	73*
GOALCONF	Confidence in own ability to achieve financial goals	.	-57*
PRODHAVE_4	Retirement Account (such as a 401k or IRA)	.	-42
PRODHAVE_5	Pension	.	.
PROPPLAN_3	I set financial goals for what I want to achieve with my money	91*	.
PROPPLAN_4	I prepare a clear plan of action w/ detailed steps to achieve my financial goals	73*	.
SELFCONTROL_3	I am able to work diligently toward long-term goals	.	.

Printed values are multiplied by 100 and rounded to the nearest integer. Values greater than 0.5 are flagged with an ". Values less than 0.35 are not printed.

From the factor pattern table we see how factor 1 significantly correlates with setting goals and planning how to achieve said goals. The signs of the correlations are in accordance with our expectations, such that respondents with current financial goals have a detailed plan to achieve it. This partial factor of FF is therefore interpreted as *financial planning (FPL)*. Factor 2 significantly and positively correlates with the two survey questions concerning FS along with confidence in ability to achieve goals, and whether the respondent has a retirement account. High confidence correlates negatively

³The Supplemental Nutrition Assistance Program (SNAP) provides a monthly benefit to buy foods for low-income U.S. citizens.

with feelings of being behind with finances. We interpret this partial factor of FF as *financial confidence (FCONF)*.

The two factors are negatively correlated (-40). A greater tendency engage in *financial planning* correlates with higher levels of *financial confidence* (due to the reverse coding of some questions). Respondents who make plans and stick to them will be more confident in their ability to achieve these goals and be less pessimistic about their future financial freedom (Barker, 2018). We note that having a pension and self control are not fitted to a factor.

We continue by identifying common factors for the variables we believe match the constituent, *future financial security (FS)*. We find only 1 eigenvalue greater than 1, but choose to estimate a model with 2 factors due to the second eigenvalue being 0.74, which is significantly higher than the remaining potential factors. Furthermore, describing each dimension with two factors eases comparison and intuitive understanding in later models. We look at the factor pattern in table 3.5.

Table (3.5) Factor pattern for FS

Variable	Label	Factor 1	Factor 2
ABSORBSHOCK	Confidence in ability to raise \$2,000 in 30 days	63*	.
FWB1_1	I could handle a major unexpected expense	88*	.
FWB1_2	I am securing my financial future	80*	.
PRODHAVE_1	Checking or Savings Account at a bank or credit union	.	40
PRODHAVE_2	Life Insurance	.	50
PRODHAVE_3	Health Insurance	.	65*
PRODHAVE_6	Non-Retirement Investments (such as stocks, bonds or mutual funds)	39	.
SAVEHABIT	Putting money into savings is a habit for me	62*	.
VOLATILITY	Household income volatility	.	.

Printed values are multiplied by 100 and rounded to the nearest integer. Values greater than 0.5 are flagged with an "*". Values less than 0.35 are not printed.

From the factor pattern table we see how factor 1 significantly correlates with the two survey questions concerning FS along with confidence in ability to raise \$ 2,000 in 30 days, exhibiting saving habits and investing in stocks, bonds or mutual funds. The signs of the correlations are in accordance with our expectations. Respondents who feel they can handle a major unexpected expense also exhibit greater tendency to save. This partial factor of FS is therefore interpreted as *unexpected expenses (UEX)*. Factor 2 significantly and positively correlates with having life and health insurance along with owning a savings account why we interpret this factor as *unexpected emergencies (UEM)*.

The two factors are positively correlated (0.35). Being able to withstand *unexpected expenses* correlates positively with ability to handle *unexpected emergencies*. Respondents who can afford insurance are likely already covered when it comes to simple shocks to finances. It is possible that the correlation is not greater since most who

can afford insurance will also be more financially robust to smaller financial shocks. However, the reverse is not necessarily true. Some may only have saved for smaller unexpected expenses but cannot afford insurance. We note that household income volatility is not fitted to a factor.

We continue by identifying common factors for the variables we believe match the presented definition of *mental health (MH)*. We find 1 eigenvalue greater than 1 and therefore only fit a single factor to the variables. We look at the factor pattern.

Table (3.6) Factor pattern for MH

Variable	Label	Factor 1
SWB_1	I am satisfied with my life	79*
SWB_2	I am optimistic about my future	83*
SWB_3	If I work hard today, I will be more successful in the future	62*
CHANGEABLE	Belief that ability to manage money is NOT changeable	.
DISTRESS	Lot of stress in respondent's life	-39
GOALCONF	Confidence in own ability to achieve financial goals	46
HEALTH	In general, would you say your health is . . .	45
MEMLOSS	Experienced memory loss or confusion in past 12 months	.
SHOCKS_11	Provided unexpected financial support to a family member or friend	.

Printed values are multiplied by 100 and rounded to the nearest integer. Values greater than 0.5 are flagged with an ". Values less than 0.35 are not printed.

From the factor pattern table we see the factor significantly correlates with the three subjective well-being (SWB) survey items along with confidence in ability to achieve goals, perceived health and stress level. All variables positively correlate with the factor except for stress level which is reverse coded, such that higher levels of stress correlates negatively with lower health, lower SWB etc. We note that belief in changeable money management ability, providing financial aid for a close relation and experiences with memory loss is not fitted to this factor. We denote this factor *mental health*.

Of the four constituents of mental health, as defined by WHO, this factor is a satisfactory measurable approximation. However, important distinctions from the definition exists. First, the subjective well-being survey items are proxies for ability to realize potential. As mentioned earlier, this is one of the most difficult measures to obtain objectively. Second, level of stress is a proxy for ability to cope with the normal stresses of life. It is evident that these two are not equal. One respondent may have high levels of stress but be proficient in coping with it, whilst another may have low levels of stress but not be able to cope. However, since this variable negatively correlates with life satisfaction and optimism, the ability to cope may be part of the variation of the factor. Ability to work productively is estimated by perceived general health and the belief that hard work will pay off in the future. Lastly, ability to contribute to community was supposed to be measured by providing financial support for a relation,

but as described, this variable did not correlate significantly with the factor. As such, one of the four constituents of mental health is not included in the factor.

3.3.2 Preliminary confirmatory factor analysis

In order to assess whether inter-factorial covariance exists between the constituents of financial well-being and mental health, we estimate a confirmatory factor model. We define the factors found in the previous section and look at the standardized covariances among factors.

Table (3.7) Standardized covariances for factors

Var1	Var2	Parameter	Estimate	Standard Error	t-Value	Pr > t
JoF	Mat	cf1f2	-0.094	0.017	-5.6	<.0001
JoF	FCONT	cf1f3	-0.985	0.005	-202.9	<.0001
JoF	Bud	cf1f4	0.126	0.017	7.36	<.0001
JoF	FPL	cf1f5	0.325	0.016	20.3	<.0001
JoF	FCONF	cf1f6	-1.038	0.006	173.2	<.0001
JoF	UEX	cf1f7	0.967	0.005	181.7	<.0001
JoF	UEM	cf1f8	0.449	0.014	30.5	<.0001
JoF	MH	cf1f9	-0.617	0.011	-55.3	<.0001
Mat	FCONT	cf2f3	0.085	0.016	5.3	<.0001
Mat	Bud	cf2f4	-0.046	0.016	-2.9	0.003
Mat	FPL	cf2f5	-0.004	0.015	-0.2	0.817
Mat	FCONF	cf2f6	0.132	0.018	7.14	<.0001
Mat	UEX	cf2f7	-0.041	0.015	-2.7	<.0001
Mat	UEM	cf2f8	-0.098	0.020	5.0	<.0001
Mat	MH	cf2f9	0.010	0.015	0.7	0.4997
FCONT	Bud	cf3f4	-0.180	0.016	-11.2	<.0001
FCONT	FPL	cf3f5	-0.362	0.015	-24.2	<.0001
FCONT	FCONF	cf3f6	1.032	0.005	210.8	<.0001
FCONT	UEX	cf3f7	-0.961	0.004	226.1	<.0001
FCONT	UEM	cf3f8	-0.474	0.013	-35.2	<.0001
FCONT	MH	cf3f9	0.520	0.012	44.6	<.0001
Bud	FPL	cf4f5	0.866	0.009	98.6	<.0001
Bud	FCONF	cf4f6	-0.190	0.017	-10.9	<.0001
Bud	UEX	cf4f7	0.186	0.016	11.4	<.0001
Bud	UEM	cf4f8	0.015	0.020	0.7	0.47
Bud	MH	cf4f9	-0.211	0.016	-12.9	<.0001
FPL	FCONF	cf5f6	-0.396	0.016	-25.1	<.0001
FPL	UEX	cf5f7	0.418	0.014	28.9	<.0001
FPL	UEM	cf5f8	0.116	0.020	5.7	<.0001
FPL	MH	cf5f9	-0.340	0.015	-22.1	<.0001
FCONF	UEX	cf6f7	-0.951	0.006	151.6	<.0001
FCONF	UEM	cf6f8	-0.556	0.010	-53.5	<.0001
FCONF	MH	cf6f9	0.596	.012	49.5	<.0001
UEX	UEM	cf7f8	0.482	0.014	33.6	<.0001
UEX	MH	cf7f9	0.556	0.010	-53.5	<.0001
UEM	MH	cf8f9	-0.226	0.012	11.6	<.0001

Note: Correlations numerically greater than 1 are discussed in later chapters.

Table 3.7 serves as evidence to suggest that our produced factors correlate significantly.

Few exceptions exist, those being no significant correlation between

1. materialism and financial planning (Mat and FPL)
2. materialism and mental health (Mat and MH)
3. budgeting and unexpected emergencies (Bud and UEM)

The lack of significant correlation between *materialism* and *financial planning* may be due to awareness of your own level of materialism. If you are good at planning you likely take your level of materialism into account, no matter the magnitude. The second independence is somewhat surprising. We assumed that higher levels of materialism might promote lower levels of *mental health*. We assumed that being inclined to impress others with your material possessions would have a negative impact on your levels of stress and community contribution. We hypothesize that the potential negatives of materialism may be offset by the potential positives. If materialism necessitates greater wealth production, this may lead to increases in productivity - a constituent of mental health.

In addition, some materialists may receive positive feedback on behalf of their material possessions, and this positive feedback may be more easily found due to the emergence of social media. However, we notice that materialism still correlates with *joy of freedom*, which in turn correlates with mental health, establishing the possibility for an implicit connection. We note the lack of significance between materialism and mental health and expect it to continue throughout the analysis. The last insignificant term is expected. A budget rarely takes *unexpected emergencies* into account which is why having health insurance etc. is uncorrelated with the ability to budget.

We use the confirmatory factor analysis as a guide for our further analysis. If most factors had not been correlated, further investigation of the relationship between financial well-being and mental health would be futile. In order to compartmentalize perceived financial well-being, we have to impose that constituents are significantly correlated. If they are not, we cannot assume that they jointly describe a common phenomenon. Moreover, the purpose is to assess the correlation between the constituents of financial well-being and mental health. The confirmatory factor analysis is therefore used as evidence to suggest significant correlation.

3.4 Scoring the original dataset

From the explorative factor analysis, we save the estimated standardized scored coefficients of the factors for use in further analysis. This enables scoring onto the original dataset or even the construction of a completely new dataset that has not been used

for estimation. We score the original dataset using the 9 factors presented earlier. This results in a new dataset with (again) 5,511 observations and additional 9 variables. As such, each observation in the original dataset is scored by each individual factor. This enables further, simplified analysis with e.g. linear regression, where only the defined factors are used as variables. We modify each factor scoring, such that the interpretation is coherent for each factor. When interpreting the factor scores, a high value reflects a positive financial well-being trait, and likewise a low value indicates a negative financial well-being trait, i.e poor financial control.

4 Model selection and estimation

4.1 Structural equation models

With constructed factors encapsulating each dimension within the definition of mental health and perceived financial well-being, we examine proposed causal paths. The structural equation modeling framework is beneficiary in supporting a proposed causal connection between latent variables. In the following section, we propose four different models, each corresponding to a different correlation path between the nine previously defined latent factors.

After constructing the initial model, it is possible to make further improvements by analyzing modification indexes, which calculates and suggests the covariances that contributes the least to the overall model. These modifications could then manually be incorporated, for instance by setting the covariance between two latent factors to zero.

When analyzing and comparing the following four models, we however choose not to incorporate modification feedback, but instead compare the baseline models where all covariances, standard deviations and residual terms are defined in accordance with our original expectations. We choose this approach primarily based on two premises. Firstly, the amount of modifications proposed in a model with 9 latent variables, each defined by several manifest variables would be excessive, and a maximum number of modifications would therefore need to be implemented, such that we compare models on an equal basis. Secondly, we are not interested, nor is it the purpose to find the best numerical SEM-model, but instead to propose the correlation path, which seems to be most statistically realistic, within the framework.

Furthermore, numerical complications caused certain eigenvalues and residuals to become negative. This is calibrated by manually defining bounds for the linear equations and residual variances. These complications are purely numerical and do not change the overall results from the models presented. The following four models are visualized in figure 4.1 and summarized in table 4.1 below.

The first SEM-model consists of a single correlation path where mental health is expected to define the remaining 8 latent factors. This model clarifies whether evidence exists for individual financial well-being being explained by their mental health. This entails that each dimension of financial well-being is a product of the corresponding mental health of the individual. This model is rooted in the belief that many parts of individual behavior reflects their mental health. Therefore, every aspect within present or future financial planning, status, freedom or security can be traced back to and be

Figure (4.1) Proposed causal paths

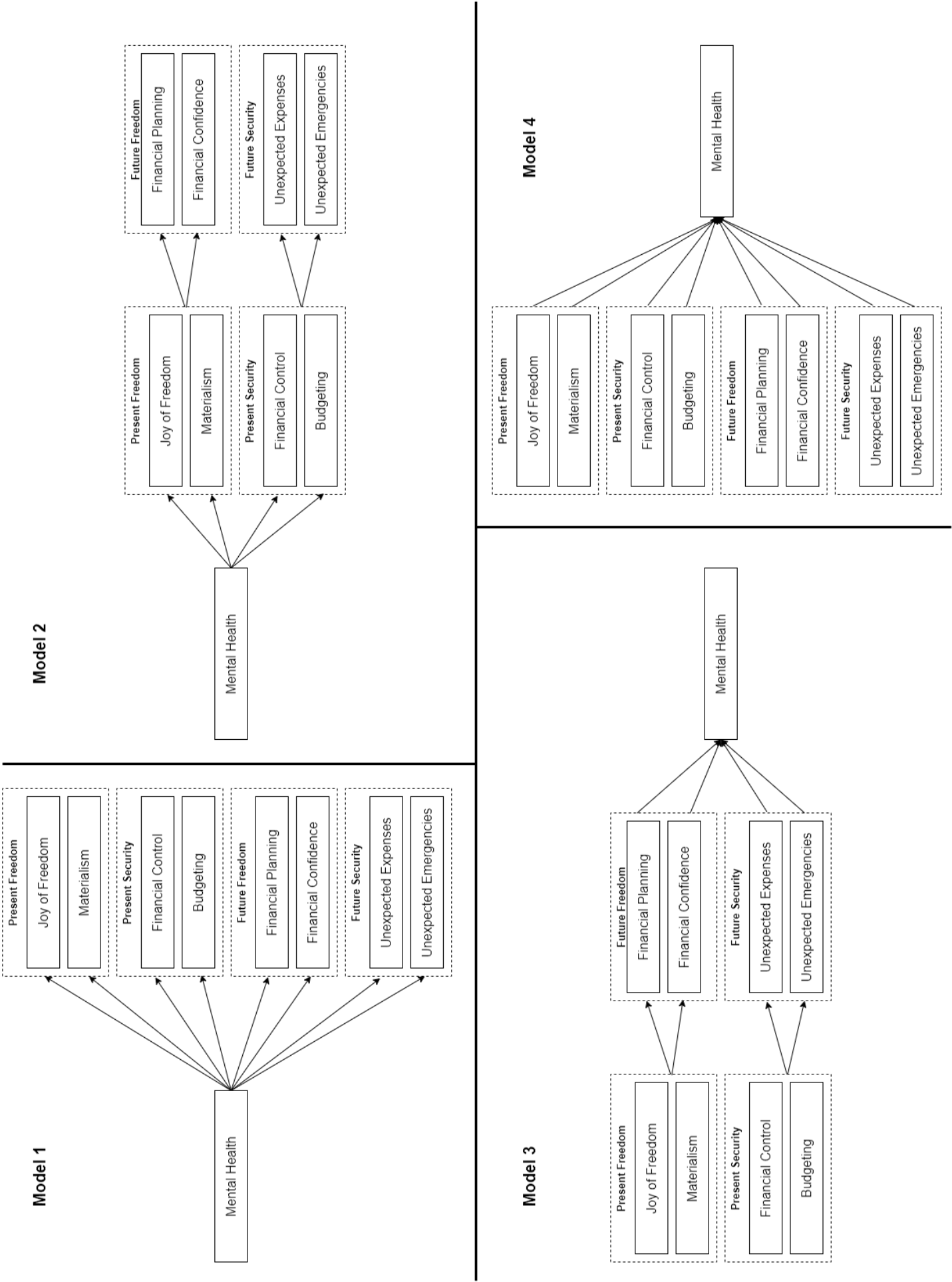


Table (4.1) SEM - models

	Models			
	Model 1	Model 2	Model 3	Model 4
Paths				
<i>Mental health</i> → <i>Joy of freedom</i>	0.50***	0.50***		
<i>Mental health</i> → <i>Materialism</i>	0.09***	0.07***		
<i>Mental health</i> → <i>Financial control</i>	−1.00***	−1.00***		
<i>Mental health</i> → <i>Budgeting</i>	−0.21***	−0.24***		
<i>Mental health</i> → <i>Financial planning</i>	−0.29***			
<i>Mental health</i> → <i>Financial confidence</i>	0.50***			
<i>Mental health</i> → <i>Unexpected expenses</i>	−0.88***			
<i>Mental health</i> → <i>Unexpected emergencies</i>	−0.12***			
<i>Joy of freedom</i> → <i>Mental health</i>				0.50
<i>Materialism</i> → <i>Mental health</i>				−0.02*
<i>Financial control</i> → <i>Mental health</i>				0.34***
<i>Budgeting</i> → <i>Mental health</i>				−0.03
<i>Financial planning</i> → <i>Mental health</i>			−0.07***	−0.06
<i>Financial confidence</i> → <i>Mental health</i>			0.09*	0.09
<i>Unexpected expenses</i> → <i>Mental health</i>			−0.23***	−0.15***
<i>Unexpected emergencies</i> → <i>Mental health</i>			0.09**	0.04
<i>Joy of freedom</i> → <i>Financial planning</i>		−0.67***	−0.35***	
<i>Joy of freedom</i> → <i>Financial confidence</i>		1.87***	1.07***	
<i>Materialism</i> → <i>Financial planning</i>		0.04**	0.03**	
<i>Materialism</i> → <i>Financial confidence</i>		0.06***	0.03***	
<i>Financial control</i> → <i>Unexpected expenses</i>		0.92***	0.83***	
<i>Financial control</i> → <i>Unexpected emergencies</i>		0.14***	0.12***	
<i>Budgeting</i> → <i>Unexpected expenses</i>		0.01	0.02*	
<i>Budgeting</i> → <i>Unexpected emergencies</i>		−0.02***	−0.02***	
Explanatory power - R^2				
<i>Joy of freedom</i>	.	.		
<i>Materialism</i>	0.01	0.01		
<i>Financial control</i>	0.95	0.93		
<i>Budgeting</i>	0.05	0.05		
<i>Financial planning</i>	0.15	0.16	0.14	
<i>Financial confidence</i>	.	.	.	
<i>Unexpected expenses</i>	0.93	0.94	0.92	
<i>Unexpected emergencies</i>	0.16	0.17	0.17	
<i>Mental health</i>			0.38	0.43
Model fit statistics				
χ^2	23,648	22,512	17,661	13,816
DF	591	585	578	559
χ^2 / DF	40.0	38.5	30.6	24.7
RMSEA	0.08	0.08	0.07	0.07
AGFI	0.74	0.75	0.79	0.82
CFI	0.74	0.75	0.79	0.85

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

measured as a derivative of respondents mental well-being. The model is formulated mathematically below:

Linear Equations

$$\begin{aligned}
 f_{JoF} &= pf_{MH,JoF} \cdot f_{MH} + \epsilon_{JoF} \\
 f_{Mat} &= pf_{MH,Mat} \cdot f_{MH} + \epsilon_{Mat} \\
 &\dots \\
 f_{UEM} &= pf_{MH,UEm} \cdot f_{MH} + \epsilon_{UEm}
 \end{aligned}$$

std

$$\begin{aligned}
 \epsilon_{JoF}, \epsilon_{Mat}, \dots, \epsilon_{UEM} &= \mathbf{V}_{JoF}, \mathbf{V}_{Mat}, \dots \mathbf{V}_{UEM} \\
 f_{MH} &= \mathbf{V}_{MH}
 \end{aligned}$$

For simplicity, the manifest variables are omitted from the mathematical depiction of the model, but each latent factor is defined in accordance with the exploratory factor analysis. Because mental health is the only prerequisite for the remaining eight latent variables, no factor covariances need to be defined.

Estimating this SEM-model yields a χ^2 -score of 23,649, which compared with the critical values for a χ^2 -distribution with 591 degrees of freedom shows that the model is highly significant. In order for this model not to be rejected we would need to reduce the sample size to 152 observations, measured by Hoelters N. The overall model fit statistics furthermore indicate that model 1 does not provide a good fit of the data ($\chi^2/DF = 40.0$; RMSEA = 0.08; AGFI = 0.74; CFI = 0.74). It is evident that the error measure is borderline acceptable, whereas the variance and fit measures fall short. The model encapsulates 74% of the actual variance which is not sufficient for the model to be deemed acceptable. Likewise the fit or interpretation statistic, CFI is below the threshold of 0.9. For the following models, the magnitude of these statistics will not be described in detail, but briefly compared.

Several paths would normally exceed the bounds without manual limitations, and thus not all paths are statistically significant. Mental health is shown not to be a significant determinant of *Joy of freedom*, *Financial control* and *Financial confidence*.

The model supports correlation paths where mental health is a determinant of *Materialism* ($\beta = 0.09, p < .0001$), *Budgeting* ($\beta = -0.21, p < .0001$), *Financial planning* ($\beta = -0.29, p < .0001$), *Unexpected expenses* ($\beta = -0.88, p < .0001$) and *Unexpected emergencies* ($\beta = -0.12, p < .0001$). When analyzing how much variance *Mental health* as a determinant explains, we obtain R^2 -values between 0.006 – 0.95. Model 1 explained 0.6% of the variance in *Materialism* and 95.0% of the variance in *Unexpected expenses*. Considering fit statistics, insignificant correlation paths and relatively low R^2 for certain latent factors, we deem model 1 not to be the optimal path. Even though arguments could be made for *Mental health* being a determinant for certain aspects of financial well-being, it is evident that it cannot explain every dimension.

The first alteration is presented in Model 2, where *mental health* instead predetermines the present security and freedom dimensions, which in turn are determinants for future security and freedom. The hypothesized correlation path states that *mental health* especially affects short-term behavior, and that short-term financial decisions is the primary driver in future financial well-being. Thus the link between *mental health* and financial future dimensions is indirect.

Estimating Model 2 yields a χ^2 -score of 22,512 which compared to a χ^2 distribution (585 DF) is beyond the critical values. The sample size should be reduced to $N = 158$ for the model not to be rejected. Fit statistics further indicate that Model 2 does not provides a good fit of the data ($\chi^2 / DF = 38.5$; RMSEA = 0.08; AGFI = 0.75; CFI = 0.75). Eventhough the correlation paths are altered, mental health is still shown to be a poor determinant for certain latent factors. *Budgeting* as a determinant for *unexpected expenses* is likewise shown not to be significant ($\beta = 0.01, p < 0.47$). Otherwise, all new correlation paths between present and future dimension are statistically significant, as seen in table 4.1. Variance explained by model 2 varies between R^2 -values from 0.01 - 0.94, where 0.6% of the variance in *Materialism* and 93.7% in *Unexpected expenses* are explained. Model 2 does perform relatively better than model 1 when measuring χ^2 -score, model fit statistics and variable estimates, but model 2 is far from acceptable. However, the model provides evidence to suggest that present financial well-being determines future financial well-being. The following model will thus shift the proposed causality, where *mental health* no longer is a determinant for financial well-being.

Model 3 proposes a causal relationship where present financial well-being still determines future financial well-being, which further determines mental health. Intuitively, this correlation path states that individuals firstly notice present financial scenarios,

which affect their future financial position which was proposed in model 2. In this model, mental health is directly determined by future financial well-being and indirectly determined by present financial well-being. This correlation path is more consistent with the hypothesis where financial well-being affect individuals mental health, rather than mental health determining financial well-being.

Model 3 yields a χ^2 -score of 17,661, which is lower than the preceding models, but still beyond the critical values for a χ^2 -distribution (578 DF). We would need to reduce the sample size to 199 for the model not to be rejected. Fit statistics for the model further indicate that this model is an improvement, compared to the previous two models, but does not meet the requirements for a good fit ($\chi^2 / DF = 30.6$; RMSEA = 0.07; AGFI = 0.79; CFI = 0.79). However, model 3 does not yield any insignificant determinants, and the variance explained varies between 0.14 - 0.92, where 14% of financial planning, and 92% of unexpected expenses is explained. The interval for explained variance is narrower than the preceding models, which also supports that model 3 is preferable. The significance between present and future financial well-being in both model 2 and 3, supports our causal prediction, where present well-being is generated first, and then affects the future well-being. Future financial well-being as a determinant for mental health measures both significantly positive and negative estimates, with *Financial planning* ($\beta = -0.07, p < 0.001$), *Financial confidence* ($\beta = 0.09, p < 0.05$), *Unexpected expenses* ($\beta = -0.23, p < 0.001$) and *Unexpected emergencies* ($\beta = 0.09, p < 0.01$).

The shift in proposed causality, where mental health is explained by financial well-being, rather than being the determinant, improved the model and is thus preferable both analytically and intuitively. To further analyze the altered correlation path, we propose Model 4, which is the inverse of Model 1, where mental health is explained by all aspects of financial well-being, independent of any time dimension. The intermediate link between present and future financial well-being is therefore removed, and all 8 latent variables are proposed to explain mental health simultaneously.

Model 4 yields the lowest χ^2 -score of 13,816, which still exceeds the critical values, compared to a χ^2 -distribution (559 DF). Measured by Hoelters N, the sample size should be reduced to 246 in order for Model 4 not to be rejected. Model 4 furthermore outperforms the preceding models, when measuring fit statistics ($\chi^2 / DF = 24.7$; RMSEA = 0.07; AGFI = 0.82; CFI = 0.85). It is evident that most fit statistics are around the threshold or accepted as a good fit (Hooper et al., 2007). One drawback is seen when comparing significance of determinants. It is evident that *Joy of freedom*

($\beta = 0.50$, $p = 0.48$), *Budgeting* ($\beta = -0.03$, $p = 0.40$), *Financial planning* ($\beta = -0.06$, $p = 0.07$), *Financial confidence* ($\beta = 0.09$, $p = 0.08$) and *Unexpected emergencies* ($\beta = 0.04$, $p = 0.28$) are not significant, whereas *Materialism* ($\beta = -0.02$, $p < 0.05$), *Financial control* ($\beta = 0.34$, $p < 0.001$) and *Unexpected expenses* ($\beta = -0.15$, $p < 0.001$) are significant determinants. Model 4 increases the proportion of variance explained in mental health, such that 43% is explained.

Despite the relative large proportion of insignificant determinants, we choose Model 4 as the best proposal for a causal path. Model 4 exhibits the best fit statistics, explains much of the variance within mental health, and is in accordance with previous literature, stating that perceived financial well-being affects subjective well-being. In addition, the model is preferable due to its simplicity. As previously mentioned, the main focus with these four SEM-models was not to find the best numerical model, but the most fitting in explaining correlations for our data.

4.2 Linear regression model

We wish to assess the significance and effect of the factors concerning financial well-being on mental health. In order to identify this, we use the scored dataset with the 9 standardized factors in order to estimate a linear regression model on the data. Based on the structural equation models, we choose to estimate model 4, where mental health is the response variable and the remaining 8 factors concerning financial well-being and its constituents are explanatory variables. We estimate the following model:

$$MH_i = \beta_0 + \beta_1 PF_i^{JoF} + \beta_2 PF_i^{Mat} + \beta_3 PS_i^{FCONT} + \beta_4 PS_i^{Bud} + \beta_5 FF_i^{FPL} + \beta_6 FF_i^{FCONF} + \beta_7 FS_i^{UEX} + \beta_8 FS_i^{UEM} + \varepsilon_i, \quad (6)$$

where the variables are the abbreviation of their constituent, e.g. FS is future security and the label of the underlying factor is noted in the superscript. The factor values are coded as such that high values of the factor corresponds to higher degrees of its related label, e.g. high values for the present freedom constituent regarding *materialism* corresponds to individuals who are more materialistic. We estimate the model by Ordinary Least Squares (OLS), implementing forward selection to rank the factors based on their explanatory power and find the following.

Table (4.2) Linear regression results

Label	Estimate	Standard Error
Joy of Freedom	0.19***	0.02
Materialism	0.03*	0.01
Financial Control	-0.09***	0.02
Budgeting	0.03*	0.02
Financial Planning	0.09***	0.02
Financial Confidence	0.27***	0.02
Unexpected Expenses	0.24***	0.02
Unexpected Emergencies	-0.07***	0.01

$R^2 = 0.35$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

The parameters are standardized such that the magnitude and sign of the estimates are comparable. This model explains 35% of the variance in the mental health factor. This is slightly less than our preferred SEM model, but this is due to the constituents not being allowed to correlate in the OLS regression. It is evident that all factors are significant at the 5%-level. However, the factors regarding *materialism* ($\beta = 0.03$, $p < 0.05$) and *budgeting* ($\beta = 0.03$, $p < 0.05$) are not as statistically significant as the remaining factors, following from the SEM models. At first, we would assume that materialistic people would be more prone to worse mental health, corresponding to a negative sign on its estimate. The feeling of having to live up to the expectations of others through your material possessions may increase stress on the individual, However, materialism may not be significant for mental health since it may correlate positively with income as mentioned previously. As such, one who is more materialistic is more likely to seek higher income to satisfy their level of materialism (Chatterjee, Kumar, & Dayma, 2019). Another thing to note, is that the estimate itself is relatively small. This indicates that even though materialism is significant, it may not be relevant in explaining mental health.

Likewise, budgeting may not be a significant determinant for mental health, since sticking to a budget demands a greater degree of willpower and cognitive discipline, which might nullify its effect on mental health. The sign is however as presumed, that those who budget might have an easier time managing their finances and therefore experience less stress and therefore a greater mental health.

For the remaining highly significant factors, some signs of the estimates seem surprising at first glance. These are present security concerning *financial control* and future security concerning *unexpected expenses*. These both have negative parameter estimates. The first factor mentioned concerns ability to make ends meet ($\beta = -0.09$, $p < .001$). Why would a greater ability in making ends meet financially correlate with lower levels

of mental health? It seems counter-intuitive but perhaps one needs to take a different perspective on the problem. Those that **do** make ends meet are perhaps exposing themselves to a great deal of stress in order to do so. On the other hand, those that are unable to make ends meet may experience less stress from simply realizing their hopeless situation and giving up completely on their financial obligations. The other negative estimate is the factor concerning unexpected emergencies ($\beta = -0.07$, $p < .001$). Why would those with health and life insurance along with a savings account have lower levels of mental health than those without? It is likely that those without simply do not care as much about potential future financial troubles and as such are less stressed as individuals. It is however an interesting and surprising finding. We would believe that having insurance is supposed to reduce stress since you will not have to worry about something tragic happening to you that may force you to pay hefty medical bills. One might argue that the cost of insurance decreases disposable income to such a degree that it may affect financial well-being and therefore mental health.

The remaining signs of coefficients are positive, as expected. *Joy of Freedom* ($\beta = 0.19$, $p < .001$) relates to the feeling of financial freedom and corresponding short-term financial restraints. Hence, individuals who are not limited short-term to enjoy life, give gifts, or in other ways are in control financially, exhibit greater mental health. Intuitively, an individual is expected to feel better if present financial burdens are not apparent. Respondents with who practice *Financial Planning* ($\beta = 0.09$, $p < .001$) are likewise better off mentally.

This result may be interpreted in different ways. People who plan long-term financial goals, and follow said plan, may have a different set of personal characteristics or motivations, which gives them a higher probability of success throughout their lifespan. Contrarily, one could argue that wealthy individuals are more likely to set long-term goals, either independently or through a financial advisor. Respondents who exhibit higher *Financial Confidence* ($\beta = 0.27$, $p < .001$) also have greater mental health. People who are not behind on their long-term financial plan, and are confident in their ability to meet financial goals, have fewer worries and restraints, other things equal. Lastly, respondents who can withstand *Unexpected Expenses* ($\beta = 0.24$, $p < .001$) do not have the same pressure and fear of a bad event. An unexpected expense of a certain proportion could be detrimental to individuals without accumulated savings or a stable income level. It is however notable that the sign of the two factors regarding future unexpectancy have different signs.

We identify the factors with greatest explanatory power by their partial contribution to R^2 . The following table illustrates the factors contributing most in explaining mental health, sorted from greatest effect to smallest:

Table (4.3) Selection Summary: Partial R^2

Step	Label	Partial R-Square	Model R-Square
1	Financial Confidence	0.295***	0.295
2	Unexpected Expenses	0.032***	0.327
3	Financial Planning	0.008***	0.335
4	Joy of Freedom	0.008***	0.344
5	Unexpected Emergencies	0.004***	0.347
6	Financial Control	0.002***	0.349
7	Budgeting	0.001*	0.350
8	Materialism	0.001*	0.350

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 4.3 illustrates several key points about the data. First, the factor for *financial-financial confidence* is by far the greatest contribution to explaining the variation in mental health out of all the factors. Second, the factors concerning the future explain more of the variance in mental health than the factors concerning the present do. One might argue that it is due to (non-perfect) multicollinearity that just one of the explanatory variables represents almost the entirety of explained variance. This finding is in accordance with our assumption along with the presented findings of ambiguity aversion from the field of behavioral economics. Ambiguous future events are not preferable, and thus adds stress and weight on the mental of individuals. This is caused by individuals being more likely to expect the worst case scenario, which affects mental health accordingly (Gilboa & Schmeidler, 1989).

Generally, factors related to financial freedom seem to explain most of the variance in mental health, compared to financial security. Financial security deals with the short-term expenses and future unexpected expenses, which one might argue causes stress and lowers mental health. However, one might more easily pay insurance to cope with such eventualities than one might insure against being restricted in your freedom due to finances. Said insurances may prevent individuals from experiencing the same level of stress related financial security, thus reducing the magnitude. In addition, one might argue that financial freedom is simply more important than financial security. Financial security is a prerequisite for upholding financial freedom and not the other way around.

When put into the context of ambiguity, financial confidence as the primary contributor fits well in explaining mental health. Since financial confidence relates

to questions about long-term financial goals, and retirement savings, this naturally involves much uncertainty. The path towards a financial goal is lengthy and many unobserved or unpredictable events can happen to the individual. If one were to fully insure themselves towards every imaginable risk on the path to said financial goal, it would be both costly and exhausting. The individual cannot predict whether a fire occurs in the household, if a household member is fired, receives a permanent critical illness, or other scenarios resulting in potential financial distress. Each event may be independently insurable, but most individuals cannot account for everything.

Furthermore, for most people, retirement savings are the primary tool to ensure a certain quality of life, when the working age has passed. It is therefore essential to secure some form of income level through some type of retirement saving. It is not, however a systemic standard for American citizens to have a retirement account, such as a 401k. In our data sample, 42.4% of respondents have no retirement account. It is questionable why such a large proportion of respondents lack a fundamental way to maintain income over their lifespan, especially given its considerable effect on mental health. This will be discussed further, later on. On the contrary, it is not difficult to understand why a lack of retirement saving can cause financial stress, and furthermore affect mental health. The majority of respondents who have a retirement saving are also exposed to financial stress, in the form of stock market deviations. This entails that whether or not one saves for retirement, some level of stress is still present. Combining the ambiguous future of financial goals, and freedom related to retirement, it becomes apparent why this factor affects mental health.

4.3 The relevance of subjectivity

From the literature review, we observed that recent studies acknowledged the relevancy of subjective measures. In the factor analysis and regression model, we include a combination of objective and subjective measures. The reason behind this choice is that we account for as many aspects of financial well-being as possible. Thus, we are able to formulate a model that includes personal thoughts or judgments in addition to traditional objective measures. In addition, one might argue that simple objective conclusions on the financial well-being of individuals do not change their perception. If you are deemed to have great financial well-being but feel like your financial well-being is low, how does it help you? To challenge the statement set by (Diener, 1984), we conduct a similar analysis, but where only objective indicators are used. If subjective measures are relevant and important in understanding both mental health and financial well-being, we would expect the objective model to be significantly inferior.

Firstly, we define dimensional factors within the same framework as in figure 1.1, where we afterwards remove insignificant contributors. Then, we score the new objective factors by following the same procedure as in section 3.4. Lastly, we estimate the following modified version of equation (6):

$$MH_i = \beta_0 + \beta_1 PF_i^o + \beta_2 PS_i^o + \beta_3 FF_i^o + \beta_4 FS_i^o + \epsilon_i, \quad (7)$$

where the superscript *o* indicates that the factor only includes objective measures. Estimating the objective regression from (7) yields the following results:

Table (4.4) Objective regression results

Label	Estimate	Standard Error
Present Freedom (BENEFITS_5)	0.10***	0.02
Present Security	0.31***	0.01
Financial Freedom	0.23***	0.02
Future Security	0.03**	0.02
$R^2 = 0.15$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$		

Table (4.5) Objective Regression: Partial R^2

Step	Label	Partial R-Square	Model R-Square
1	Present Security	0.111***	0.111
2	Future Freedom	0.0380***	0.149
3	Present Freedom (BENEFITS_5)	0.003***	0.153
4	Future Security	0.001**	0.153
* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$			

Ruling out variables concerning subjective perception leads to significantly less explained variation for the mental health factor. The decrease in R^2 is therefore not just due to fewer variables making up the factors, even though it is a contributing factor. This serves as evidence to suggest that *perceived* financial well-being is a crucial addition to understanding both financial well-being itself and also its relation to mental health. Modeling financial well-being from a purely objective position omits a crucial part of its correlation. A gross simplification is to say that about half of the total variation in mental health, that is explained by financial well-being, consists of objective measures and the remainder consists of individual perception.

Furthermore, the partial R^2 is different for factors compared to the previous regression where both objective and subjective variables are used. Following the forward selection steps from the objective regression, we find that the *Present Security* factor accounts for the majority of explained variance. This is in direct contradiction to the first linear regression, where *Present Security* was among the least contributing factors

to explaining variance in mental health. This presents an interesting finding not yet discussed in the literature: certain elements of financial well-being may be more easily explained by objective measures compared to others. Financial present security is more easily measured objectively through duly bill payments, debt collector interaction and snap benefit support. Whether individuals perceive themselves to be in control of their finances may not necessarily support their measurable ability to be.

The *Future Freedom* factor remains a large contributor to overall explained variance, despite the fact that removing subjectivity from the factor decreases its explanatory power. It is evident that objective measures do not fully encapsulate the financial well-being phenomenon. Additionally, the constituents of financial well-being vary in how their explanatory power is affected by removing subjectivity from their calculation. Our results thus supports the inclusion of subjective measures when modeling financial well-being, as stated by Diener (1984).

4.4 Estimating present-bias

In order to assess the distribution of importance between present and future financial well-being on mental health, we perform a second-order factor analysis. We wish to estimate this model for intertemporal decision-making to see how respondents discount future financial well-being. The second order factor analysis provides a further inspection of future financial well-being being as the primary factor in explaining mental health. Using the defined factors, we group them into those that address the present and those that address the future. First, we examine factors relating to present financial well-being.

Table (4.6) Factor pattern for present

Variable	Label	Factor 1
PF_1	Joy of Freedom	80*
PF_2	Materialism	.
PS_1	Financial Control	100*
PS_2	Budgeting	26*

It is evident that the factor created by using the factors dealing with the present consists mostly of the correlation between the *joy of freedom* and the *financial control* factors. It is not surprising that these two factors correlate so strongly and one might argue that being in control of your finances increases your probability of being able to enjoy life, despite your limited resources. We also note that *materialism* is not contributing significantly to the factor. This result is in accordance with the SEM model linear regression model, where materialism was the least contributing and significant factor.

We repeat the procedure for factors regarding future financial well-being.

Table (4.7) Factor pattern for future

Variable	Label	Factor 1
FF_1	Financial Planning	46*
FF_2	Financial Confidence	88*
FS_1	Unexpected Expenses	87*
FS_2	Unexpected Emergencies	49*

Table 4.7 indicates that all factors regarding future financial well-being fit within the general "future" grouping. The resulting "future" factor is much more well-defined in comparison to the "present". It should be noted that the normal factor pattern threshold is lowered from 0.5 to 0.4, because both financial planning and unexpected emergencies are borderline significant. All four factors are well-represented in the created factor where *financial confidence* and *unexpected expenses* offer the most correlation. This comes to no surprise given these two factors provided the greatest partial R^2 in the linear regression model. One might argue that greater confidence increases individual robustness against the unexpected.

The broader second order factors are then analyzed by scoring the original dataset in order to estimate the following linear regression model:

$$MH_i = \beta_0 + \beta_1 present_i + \beta_2 future_i + \varepsilon_i, \quad (8)$$

where our interest lies in the magnitude of β_1 and β_2 such that we might infer the potential ambiguity aversion in individuals, as well as indicate the presence of partially naive respondents. If future financial well-being is a greater predictor of mental health, as indicated by the prior linear regression model, we expect a higher parameter estimate for β_2 along with greater significance. We utilize OLS to estimate (8), which yields the following results.

Table (4.8) Present-bias regression

Label	Estimate	Standard Error
Financial Well-Being (Present)	-0.03	0.02
Financial Well-Being (Future)	0.58***	0.02

$R^2 = 0.35$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

From the above, we learn that the second-order factor concerning present financial well-being is insignificant at explaining the variance in mental health. These results resonate with the proposed model of intertemporal decision-making, where agents use quasi-hyperbolic discounting. Individuals seem to be hurt by their propensity for present satisfaction, since it has no effect on their mental health. On the other hand, their

future financial well-being has a positive effect on their mental health. It is evident that the construction of a single factor concerning present financial well-being, such that it may have a measurable effect on mental health, is difficult. The main takeaway from the model is that $\beta_2 = 0.58$, is relatively large, thus in accordance with our expectations. This further supports our findings from the previous linear regression model, indicating that future financial well-being is the most important determinant for mental health of the two time dimensions. In addition, it follows our assumption that individuals are hurt by their preference for immediate financial freedom and security rather than also preferring that of the future. Despite an insignificant "present" factor, we conclude that this model provides evidence to suggest that future financial well-being is weighted higher with regards to individual mental health, and individuals exhibit naive behavior concerning the trade-off between present and future well-being.

4.5 Differences between population groups

We wish to assess how factors vary across different demographic variables. These variables are presented in table 3.1. This allows for identification of potential outlier population groups in terms of mental health and financial well-being. Further, this allows financial advisors and policy makers to tailor better suggestions and policies towards different parts of the population. We wish to examine how *mental health*, *financial confidence* and *joy of freedom* vary across these control variables and whether noticeable differences exist between categories. We look at these factors specifically, since they contribute with the majority of explained variance in mental health. Formally, we test the hypotheses:

$$H_0 : LS\text{Mean}(i) = LS\text{Mean}(j) \quad (9)$$

where $i \neq j$ and LSMean is the least squares mean for each categorical variable i, j . The parameter estimates are least squares means which complicates intuitive interpretation. However, it allows for an easy comparison of magnitude and statistical significance differences between categories. We estimate a generalized linear model (GLM) in order to conduct t-tests for significant differences. An overview of the conditional means is presented in table 4.9 below.

Focusing on the *mental health* factor, our findings is in accordance with the proposed theory of the "U-shape of happiness" (Cheng, Powdthavee, & Oswald, 2017). This empirically supported theory, describes how people are most happy at the beginning and end of their lives, and less happy in the ages where they have entered the workforce and start having families. The pre-boomer (-0.02) and millennial generations (-0.08)

Table (4.9) Conditional LS-means

Control variable	Factor								
	PF		PS		FF		FS		MH
	JoF	Mat	FCONT	Bud	FPL	FCONF	UEX	UEM	MH
Generation									
<i>Pre-Boomer</i>	0.18	0.06	0.20	0.11	-0.08	0.10	0.15	-0.13	-0.02
<i>Boomer</i>	-0.11	-0.06	-0.14	0.08	-0.04	-0.15	-0.14	-0.19	-0.19
<i>Gen X</i>	-0.37	0.04	-0.45	0.01	-0.12	-0.40	-0.39	-0.28	-0.21
<i>Millenial</i>	-0.40	0.18	-0.44	0.04	0.07	-0.37	-0.41	-0.41	-0.08
Gender									
<i>Male</i>	-0.15	0.11	-0.18	0.04	-0.03	-0.21	-0.14	-0.26	-0.14
<i>Female</i>	-0.20	0.00	-0.23	0.08	-0.04	-0.20	-0.25	-0.24	-0.10
Education									
<i>Less than high school</i>	-0.35	0.14	-0.50	-0.01	-0.23	-0.52	-0.53	-0.60	-0.33
<i>High school degree/GED</i>	-0.18	0.09	-0.24	0.02	-0.10	-0.23	-0.27	-0.31	-0.15
<i>Some college/Associate</i>	-0.22	0.06	-0.23	0.09	-0.01	-0.20	-0.23	-0.22	-0.09
<i>Bachelor's degree</i>	-0.09	0.01	-0.05	0.09	0.05	-0.04	0.00	-0.06	-0.06
<i>Graduate/professional degree</i>	-0.04	-0.01	0.01	0.10	0.09	-0.03	0.04	-0.06	-0.00
Ethnicity									
<i>White, Non-hispanic</i>	-0.16	0.00	-0.13	0.07	-0.15	-0.16	-0.15	-0.09	-0.19
<i>Black, Non-hispanic</i>	-0.17	0.07	-0.40	0.07	0.09	- 0.27	-0.27	-0.24	-0.09
<i>Other, Non-hispanic</i>	-0.29	0.07	-0.19	0.02	-0.18	-0.29	-0.25	-0.24	-0.36
<i>Hispanic</i>	-0.09	0.10	-0.10	0.21	0.09	-0.11	-0.12	-0.42	0.13
Marital status									
<i>Married</i>	-0.05	0.02	-0.12	0.12	0.07	-0.08	-0.11	-0.12	0.02
<i>Widowed</i>	-0.15	0.03	-0.17	0.20	0.02	-0.11	-0.09	-0.11	-0.01
<i>Divorced/Separated</i>	-0.25	0.05	-0.28	0.05	-0.02	-0.26	-0.28	-0.27	-0.19
<i>Never married</i>	-0.15	0.13	-0.15	-0.09	-0.16	-0.23	-0.22	-0.39	-0.26
<i>Living with partner</i>	-0.27	0.06	-0.30	0.01	-0.11	-0.35	-0.28	-0.35	-0.18
Income									
<i>Less than \$20,000</i>	-0.67	0.11	-0.78	0.12	-0.13	-0.69	-0.69	-0.80	-0.52
<i>\$20,000 to \$29,999</i>	-0.50	-0.05	-0.54	0.02	-0.14	-0.53	-0.59	-0.49	-0.40
<i>\$30,000 to \$39,999</i>	-0.44	0.08	-0.44	0.07	-0.18	-0.43	-0.50	-0.37	-0.35
<i>\$40,000 to \$49,999</i>	-0.24	-0.01	-0.26	0.11	-0.04	-0.26	-0.30	-0.25	-0.04
<i>\$50,000 to \$59,999</i>	-0.16	-0.03	-0.19	0.09	-0.08	-0.15	-0.16	-0.12	-0.01
<i>\$60,000 to \$74,999</i>	-0.05	0.02	-0.05	0.10	0.02	-0.07	-0.04	-0.08	-0.01
<i>\$75,000 to \$99,999</i>	0.00	0.12	-0.02	-0.01	0.00	-0.04	-0.01	-0.11	-0.01
<i>\$100,000 to \$149,999</i>	0.14	0.10	0.13	0.02	0.03	0.07	0.15	-0.04	0.02
<i>\$150,000 or more</i>	0.33	0.18	0.31	-0.01	0.14	0.25	0.38	0.00	0.23

exhibit highest mental health with no significant difference between them but both groups have significantly higher mental health than boomers (-0.19) and gen x's (-0.21) who have the same (lower) level of mental health. We also find no significant difference in mental health between male (-0.14) and female (-0.10) respondents.

Mental health seems to be increasing with level of education. A single increase in education level (eg. going from high school to some college) is not related to a significant change in mental health. However, two steps of educational level increase corresponds to significantly greater mental health. This is an important distinction to other variables, since education is alterable, whereas one cannot change one's date of birth, biological gender and ethnicity. The positive relationship between educational degree and mental health could be caused by more educated individuals accumulating more financial knowledge and monetary success than their lesser educated counterparts. Therefore, educated individuals are more financially stable and exhibit greater financial well-being, which was previously shown to correlate with higher levels of mental health. However, we note that the reverse causality is plausible.

Mental health is significantly different between all ethnic groups. In order of lowest to highest mental health, the "other, non-hispanics" group (-0.36) exhibit the lowest mental health, followed by whites (-0.19), then blacks (-0.09) and with the highest mental health resides the hispanics (0.13). Cultural differences between ethnicities may help with understanding why some do better than others, along with identifying what to do and what not to do in order to increase mental health.

Married (0.02) or widowed (-0.01) respondents exhibit significantly greater mental health than those that are divorced (-0.19), never married (-0.26) or simply living with a partner (-0.18). We note that these variables are correlated such that women are over-represented in the widows category. These correlations may explain some of the differences. Finally, income is almost perfectly linear in mental health with the highest earning respondents exhibiting the highest degree of mental health. Notably, education level and income are correlated, but not perfectly. Therefore, having low levels of education, but a high income also leads to greater mental health. Mental health may be improved both by simply being educated but also in the way of increased income.

We move on to *financial confidence* which provided the greatest degree of explanatory power in our linear regression model. The relationship between financial confidence and generation seems convex. Gen X's (-0.40) and Millenials X's (-0.37) have equally low financial confidence and boomers (-0.15) have significantly greater confidence exceeded by pre-boomers (0.10). This comes as no surprise since the elderly may be less concerned

about the things they want in life, since most of it has been lived. They are also more likely to have a retirement account and due to a possibly longer working life, their confidence in raising money is higher.

Just as before, we find no difference between genders. The relationship between education level and financial confidence is concave with a greater increase going from less than high school (-0.52) to a high school degree (-0.23) than going from a bachelor's degree (-0.04) to a graduate degree (-0.03). However, financial confidence is increasing in education level which is a reason to take an education in the first place; to make money in order to have a less stressful financial future. With regards to ethnicity, whites (-0.16) and hispanics (-0.11) are equally financially confident. Less confident are blacks (-0.27) and other non-hispanics (-0.29) which exhibit an equal, yet lower financial confidence. Married (-0.08) and widowed (-0.11) respondents are significantly more financially confident than the rest. Again, these two groups are probably older on average, which as seen before indicated higher financial confidence. Furthermore, financial confidence is increasing with income.

The *joy of freedom* factor is of interest due to its comparably high parameter estimate from the linear regression model. The youngest respondents, millenials (-0.40) and gen x (-0.37) exhibit similar low levels of joy of freedom. Otherwise, the differences between generations are significant and increasing in age. Intuitively, older individuals have accumulated more wealth over their lifespan, and do not have the same short-term financial limitations as younger generations.

In contrast to the prior findings, the joy of freedom is significantly higher for males (-0.15) than for females (-0.20). This indicates the male respondents feel more in control of their present finances, are able to give gifts more easily and to a greater extent are capable of enjoying life because of their money managing. Why do males feel more financially free in the present? We do not have sufficient data to state whether inherent gender differences exist or whether behavioral differences create this discrepancy. The behavioral finance literature has found evidence to suggest greater levels of confidence in males. As shown by Barber and Odean (2001) and Glaser and Weber (2007), men exhibit a greater tendency to overestimate own abilities. This may lead them to exaggerate their well-being.

Joy of freedom with regards to education level indicates significant differences, except the highest educated individuals and between high school/college educated respondents. A positive trend is present, meaning that the higher educated do not have the same constraints, and are able to enjoy short-term financial freedom, relative to

lesser educated respondents. As mentioned previously, this is likely because of income effects related to increased education. Another difference of interest is in ethnicities, where other, non-hispanics (-0.29) exhibit significantly less joy of freedom than the rest. Married people (-0.05) have significantly higher joy of freedom, than the other marital groups. This is likely because married people have more than one source of income, and share expenses, which enable them to enjoy a higher disposable income, other things equal, further enabling more short-term financial freedom. Again, joy of freedom is significantly increasing in income.

Of the remaning factors, we note the following, interesting significant differences. With regards to *materialism*, men are more materialistic than women and materialism decreases in education level. Blacks score significantly lower on *financial control* than other ethnicities. Men are better at handling *unexpected expenses* than women, but handle *unexpected emergencies* equally well. The older generations handle unexpected costs better than the young generations and higher income increases your capacity to absorb financial shocks.

Significant differences between population groups exist with regard to mental health and financial well-being. These differences highlight the varying presence of the correlation between mental health and financial well-being amidst the population. These findings may support increasingly tailored solutions for consumers.

5 Discussion, interpretation and application

The study demonstrates a significant, positive correlation between perceived financial well-being and mental health, in line with the hypothesis. Of all the components that make up financial well-being, having *financial confidence* is by far the greatest predictor of high mental health. This implies that being confident in your ability to reach your financial goals, having a retirement account, being on track with your finances and feeling that your money situation allows you to have the things you want, all contribute positively to a greater mental health. From the definition of mental health, these perceptions correspond well to managing stress levels, being productive and being able to reach your potential.

We identify *financial confidence* as the most important link that connects financial well-being to mental health. Confidence may be extrapolated as the realization of the possibility that things can get better. The similarity with hope is evident and if positive change is the goal, having confidence in the possibility that the goal is achievable seems to be necessary in order to achieve it. Lastly, another interpretation is that a large portion of respondents in our data set are at least partially naive, incapable of sacrificing present utility for future utility. Naive individuals may be less likely to be confident in their future and as a result be less likely to sacrifice present utility. We expand on this proposition in more detail later on.

Our results build on existing evidence of perceived financial well-being impacting overall well-being (Netemeyer et al., 2018) (Barrett et al., 2021). The analysis provides new insights into the exact relationship between perceived financial well-being and mental health. Our results are in accordance with existing literature but contributes to the identification of the most important factors. Furthermore, our results confirm that subjective measures are important for understanding financial well-being and its relation to mental health, which is in line with existing literature. These results should be taken into account when considering how to model and compartmentalize financial well-being. While previous research focused on explaining overall well-being, we find that the majority of this explained variance likely concerns mental health, which we refined our response variable into.

The results contradict the claims of Netemeyer et al. (2018) that *materialism* is an antecedent to perceived financial well-being. It appeared as the factor with the smallest number of variables in our exploratory factor analysis. In addition, it lacked significant correlation to other factors, including mental health in both our confirmatory factor model, structural equation models and later regression. We continued to use it

in the following analyses despite of this. We assume that it does not take away from the overall model. In addition, the overall consistency remains as each constituent consists of two underlying factors. We hypothesize that the lack of significance may be explained by the difference in response variable. Materialism may explain variance in overall well-being but has no significant correlation to mental health, specifically. We also hypothesize that the potential increase in expenditure for materialistic individuals may be offset by a propensity to increase income in some way, therefore nullifying its effect on financial well-being and therefore mental health.

In line with the hypothesis, our comparison of structural equation models suggested that the causal path goes from perceived financial well-being to mental health. Netemeyer et al. (2018) drew the same conclusion based on an elaborate instrumental variable regression analysis. Determining causal paths is difficult and further research into causality is needed to further reaffirm our findings. Studying changes across time in perceived financial well-being and mental health may lead to more robust conclusions regarding causality than what is possible with our data.

5.1 Encouraging retirement saving

Financial confidence was measured to be the primary contributor to mental health. It is therefore important to understand the drivers behind this factor. This is beneficial for a variety of economic agents, such as politicians, when modeling and suggesting macroeconomic changes.

One instrument within financial confidence that can be altered by political intervention is pension and retirement saving. Specifically, these interventions should incentivize and encourage retirement savings to counteract the naive present-bias of individuals leading to lower levels of mental health. Furthermore, interventions should introduce standard retirement plans that are adapted to the individual in order to maintain consumption after exiting the workforce.

We witnessed that a relatively large proportion of respondents have no retirement plan, in the form of a traditional pension, 401k or IRA (32% of Gen X's and 43% of millennials). The lack of retirement savings is problematic for the American economy for several reasons. Firstly, individuals are unable to maintain their accustomed quality of life, and are forced to depend on social security benefits from the government. For the vast majority of people, these benefits are not sufficient to cover expenses and support similar financial well-being, as they had when they were in the workforce. Therefore, this may lead to poverty in the future. Adults of many devel-

oped countries will only manage to finance about half of their retirement life expectancy (World Economic Forum, 2019). Furthermore, retirement savings, independent of type, are included in the accumulated national savings. In recent years, the United States have continuously accumulated increased national debt, which retirement savings could contribute to reduce. In conclusion, altering the savings behavior for the general public helps counteract future poverty within a certain demographic group, and general government net lending.

It is not enough to simply have a retirement account. The retirement plan must also be adapted to the individual's economy. Therefore, the periodical deposits should be income-dependent, such that individuals can maintain their living standard. Questions about how these retirement funds are allocated within the financial markets or individual risk profiles naturally occur, but are beyond the scope of this discussion.

The American retirement system reflects the general ideological foundation of individual freedom, where no mandatory requirement is established for retirement savings. Instead, the choice is, for the most part, put in the hands of the individuals, thus presenting an intertemporal choice between present and future consumption. We commented on the potential pitfalls in having individuals be solely responsible for their own future. Many individuals cannot relate to their future selves and therefore fail to sacrifice the optimal amount of present utility for future utility. Furthermore, we found that present financial freedom does not contribute as much towards mental health as financial confidence (future financial freedom). This again serves as an argument to help individuals forego some level of present financial freedom for greater future financial freedom.

Throughout the 1980's, retirement savings partially paid by employers, such as a 401k became more widespread, but the decision to offer this was and still is up to the individual employer and therefore not required by law or union agreement. However, the responsibility for saving for retirement itself is still entirely on the individual which may resolve in insufficient retirement savings. Even if employers offer a 401K, it is not mandatory for employees to use them.

When analyzing the American retirement system, one fundamental flaw becomes apparent: The preference for freedom within American culture may hurt individuals in the long-run, because people deviate from rational behavior and tend to prioritize present consumption. We described individuals exhibiting this behavior as at least partially naive. Since no mandatory policy concerning retirement saving exist, people postpone savings, and are subjugated to their inherent present-bias, resulting in sub-optimal sav-

ing patterns. The American public is not necessarily more financially naive than other populations. However, their current policies do not protect them against their inherent behavioral inadequacies when compared to other western economies.

Looking at the lifespan periodically, savings are postponed consumption for future periods, where naive individuals still prefer current consumption, and chooses not to save for retirement. Iterating through the working age, when retirement is inevitable, retirement savings are insufficient due to this irrational and naive behavior. Policy interventions that incentivize income-dependent retirement savings could benefit a large proportion of Americans, whose naive behavior hurts them financially long-term.

The naive behavior is particularly explicit when individuals actively have to manage their retirement funds. When savings are salient, individuals have to actively resist the temptation of short-term consumption (Bordalo, Gennaioli, & Shleifer, 2013). For this reason, retirement plans that are "invisible" for the individual are preferable, since this temptation never occurs. Hence, individuals would save more efficiently if their decreased present consumption was not as salient.

These systems evidently perform comparatively well, as witnessed in countries like Denmark, The Netherlands and Sweden ((Sørensen, Billig, Lever, Menard, & Settergren, 2016)). Each of the aforementioned countries have an invisible pay-as-you-go (PAYG) structure, which is relatively similar to a mandatory 401k plan. Even though many differences between the pension systems exist between these countries, a common denominator is how the government encourages implementation of sufficient retirement plans. In Denmark for instance, the retirement agreements are decentralized, and decided between unions representing employers and employees. The government can intervene if neither party can reach an agreement. This system closely resembles a mandatory employer-matched 401k retirement plan⁴. Furthermore, institutions have been established and are heavily regulated in order to maintain said systems. Said government reliant pension system may not be the ideal solution in the United States, but a more privatized version could be proposed. A heavily incentivized or mandatory system wherein the non-salient PAYG structure is implemented could contribute to greater future financial well-being.

As stated, *financial confidence* was the major contributor in explaining mental health, which is why such a systemic change could have a large effect on the general mental health. Even though such a change may collide with classic American ideologies, the changes are relatively simple and easy to implement. In addition, the state has an

⁴A plan wherein the total contribution $T = x + s$ is split between employers who contribute $x\%$ of the employees pay check, where s is the employees own contribution.

interest in helping individuals finance their own pensions rather than having them financed by the state. Besides the individual mental health improvements, such a change could furthermore have derivative effects in other macroeconomic indicators. Further research may look into the differences in the relationship between perceived financial well-being and mental health across countries and retirement policies. We hypothesize that *financial confidence* is less important with regards to mental health in Denmark than in the US, given the different retirement schemes.

We note that retirement does not contribute as heavily to the financial confidence factor as being behind with finances does. This raises the issue that the disposable income of some individuals is low enough to only sustain an above-poverty quality of life. This results in a financial situation where future income streams are used to finance potential past bill accumulation. Some individuals do not simply face a trade-off between present utility and future utility. They face a problem where sacrificing present financial well-being results in poverty or at least insufficient living standards. As such, they have no incentive to save for retirement and would likely be worse off from a policy intervention that decreases their present income in order to increase retirement savings.

5.2 Statistical limitations

Even though much structure and care is put into the way our categorical data is constructed, ways to further improve the consistency is arguably present from a purely statistical perspective. A common practice within categorical data is to follow a likert scale, which categorizes variables into a scale of 1-5. Implementing such a conversion would enable us to utilize specific statistical measures designed for categorical data. This could for instance be the inclusion of cronbachs Alpha, that would indicate how reliable our factor groupings were. Including such reliability measures would not be the statistical backbone, but serve as an additional quality check to ensure that our chosen groupings are valid. Throughout the analysis, *materialism* performs poorly, both in terms of factor correlation and explanatory power towards mental health. A reliability check of this specific grouping utilizing cronbachs alpha could therefore identify weaknesses early in the analysis.

It is however difficult to convert all relevant variables to fit the likert scale. Binary questions such as whether the respondent has a retirement account, would be rather difficult to extend to 5 categories without reworking the question entirely.

Throughout the data cleaning process, we chose to exclude missing values, which affected the number of observations used in the final analysis. The reduction in obser-

vations was approximately 1,200, which almost equals 25% of our final data sample. We made a distinct choice to exclude the observation if the respondent chose not to answer any variables of interest. Therefore, one could argue that the model selection, estimates or proposed causality would differ, if these observations were included. Removed observations could be reintroduced into the data sample by interpolating any missing data. Instead of inserting a "best-guess" for missing values, and implementing data interpolation, we decided that the sample was of a feasible size. Furthermore, utilizing data interpolation would introduce other complications which are beyond the scope of this paper. The critical assumption behind excluding the missing values is that the missing observations are not significantly correlated with the variables used in the analysis.

When estimating whether subjective questions increase model performance, we find the objective model to be inferior. When including both subjective and objective indicators, we achieve a more concise expression for financial well-being, which is expected to explain and encapsulate more variance within mental health. Therefore, when comparing these two models, one would immediately assume that the implementation of subjective measures is necessary. This result is further in accordance with Diener (1984), where personal traits, cognitive judgment, or life experience was expected to affect financial well-being.

Within our data, some features need addressing, before the subjective model is chosen as the superior model. Firstly, the objective model from (7) is a subset of the subjective model from (6). It is therefore expected that the objective model does not explain the same amount of variance, as the subjective. If one were to add variables, the explanatory power would increase, independent of whether the variables are relevant or not.

Another shortcoming in the objective model is within the present freedom dimension. When iterating through the variable list, only one variable was deemed to be objective, meaning that the present freedom factor consists of one variable. Ideally, we wanted to generate the same factors as the subjective model, to increase the similarities. In the subjective model, factors related to present freedom did not explain much variance, and likewise we do not expect present freedom to be the main contributor in the objective model. Therefore we justify using BENEFITS_5 as a proxy for present freedom in the objective model. Comparing the subjective and objective models we see an increase in explanatory power of 20 pct. points. Including subjective measures significantly increases explained variance. Even though we did expect an increase in

explanatory power due to the inclusion of more variables, we did not expect a difference of that magnitude. The theoretical arguments for including subjective measures, combined with our empirical evidence, support the inclusion of subjective variables when explaining perceived financial well-being.

We faced a statistical issue when creating the structural equation models. For some models, we encountered non-positive definite predicted covariance matrices, even though all predicted variances for the latent variables were positive. For other models, the estimated variance of some error terms were negative. We counteracted this issue by imposing boundaries to the parameter estimates for the factors and errors. This limits exact interpretation of standard errors and χ^2 -tests due to the statistical assumption that solutions are based within the parameter space. We note how this may affect the applicability of our results, however the solution at the very least makes the results interpretable and sound in the sense that the results are within an acceptable parameter space. It is beyond the scope of this study to identify and correct the factor calculations resulting in the variance errors and attempt to remedy them.

5.3 Definitional assumptions and remarks

Throughout the factor generating process, we assign relevant variables to a specific constituent. When iterating through the variable list, for each variable we first had to assess if the variable was relevant within our framework, and second assign it to a constituent. When evaluating if a variable fits within any constituent, we went through each definition by CFPB and WHO, to eliminate as much subjective interpretation as possible. However, it is not possible to eliminate all interpretation bias, and a variable that we evaluate to affect future freedom, could maybe be interpreted by another to affect future security. It can for instance be difficult to assign PRODHAVE_1 to a single constituent. Does having a Checking or Savings Account at a bank or credit union, only fit within one constituent? We chose to assign it to Future Security, based on the argument that savings are primarily used to absorb a financial shock. It could also be interpreted as an alternative way to be on track to meet your financial goals, and therefore should be assigned to Future Freedom. These different interpretations can vary, meaning that the factor generating process include some degree of subjectivity. By using established definitions, we try to minimize said uncertainty in interpretations.

However, our explained variable, mental health, does not have a definition eases objective statistical measurement. We aimed to be more precise about the phenomenon we are viewing perceived financial well-being in relation to. We believe overall well-

being is an opaque term and mental health has gained increased attention, even more so during the pandemic. As such, it proved a fitting solution to our problem. However, simply having a definition says little of its statistical efficacy and ease of application. Establishing an academic definitonal framework from which to build upon is a crucial part of addressing the issues presented in this paper. Nevertheless, standardizing the phenomenon of interest in relation to financial well-being is still a work in progress. We proposed a possible definition but we had to subjectively pick out variables which matched the definition in order to construct a factor for the measure itself. Following research may consider using the survey approach produced by Tennant et al. (2007) in order to calculate a single measure for mental health.

6 Concluding remarks

This research aimed to identify the relationship between the constituents of perceived financial well-being and mental health. We tested the hypotheses that (H_1 :) perceived financial well-being has a significantly positive correlation to mental health, (H_2 :) that perceived financial well-being is a plausible causal determinant of mental health and (H_3 :) that perceived financial well-being regarding the future has a greater effect on mental health than that of the present.

We find that most constituents of perceived financial well-being have a significant, positive effect on mental health. Through thorough structural equation modeling, we find evidence to suggest a causal path that goes from perceived financial well-being to mental health, however these results are prone to interpretation. The factors produced to compartmentalize perceived financial well-being jointly explain 35% of the variance in mental health. The greatest effect stems from financial confidence (future freedom) and ability to mitigate unexpected expenses. Greater financial confidence and capacity to absorb unexpected expenses is associated with higher levels of mental health. We find that financial well-being concerning the future is more important to individual mental health than that of the present. A possible explanation to this is individual aversion to ambiguity which we estimate by modeling present bias. Lastly, we find a significant decrease in model performance when removing subjective variables from the factors making up each constituent, further emphasizing the importance of including subjective measures.

While the survey population limits generalizability of the results, our approach provides new insight into which parts of perceived financial well-being are most important for determining mental health and allows for further investigation of population differences. We focus on retirement savings as a part of future financial freedom, given its importance in explaining mental health. The partial naivety of individuals explains why individuals tend to save insufficient funds for retirement. This stands as an argument to support that a mandatory, less salient retirement scheme is preferable.

To better understand the implications of these results, future studies could address international differences and the effects of increasing financial confidence. Financial advisors should consider population differences when addressing financial well-being since population groups exhibit significant variation in their financial well-being constituents.

Even though financial well-being constitutes the majority of perceived individual overall well-being (Netemeyer et al., 2018), research on its specific effect on mental health is lacking. Existing literature often relies on objective measures such as income or single-item measures of overall satisfaction. However, recent research in the field has increased focus on subjectivity of financial well-being but offers a more general view of its structure and effects. This study addresses these shortcomings by more finely dividing financial well-being into its constituents and proposing a more specified effect by looking at mental health as defined by WHO instead of an overall well-being measure.

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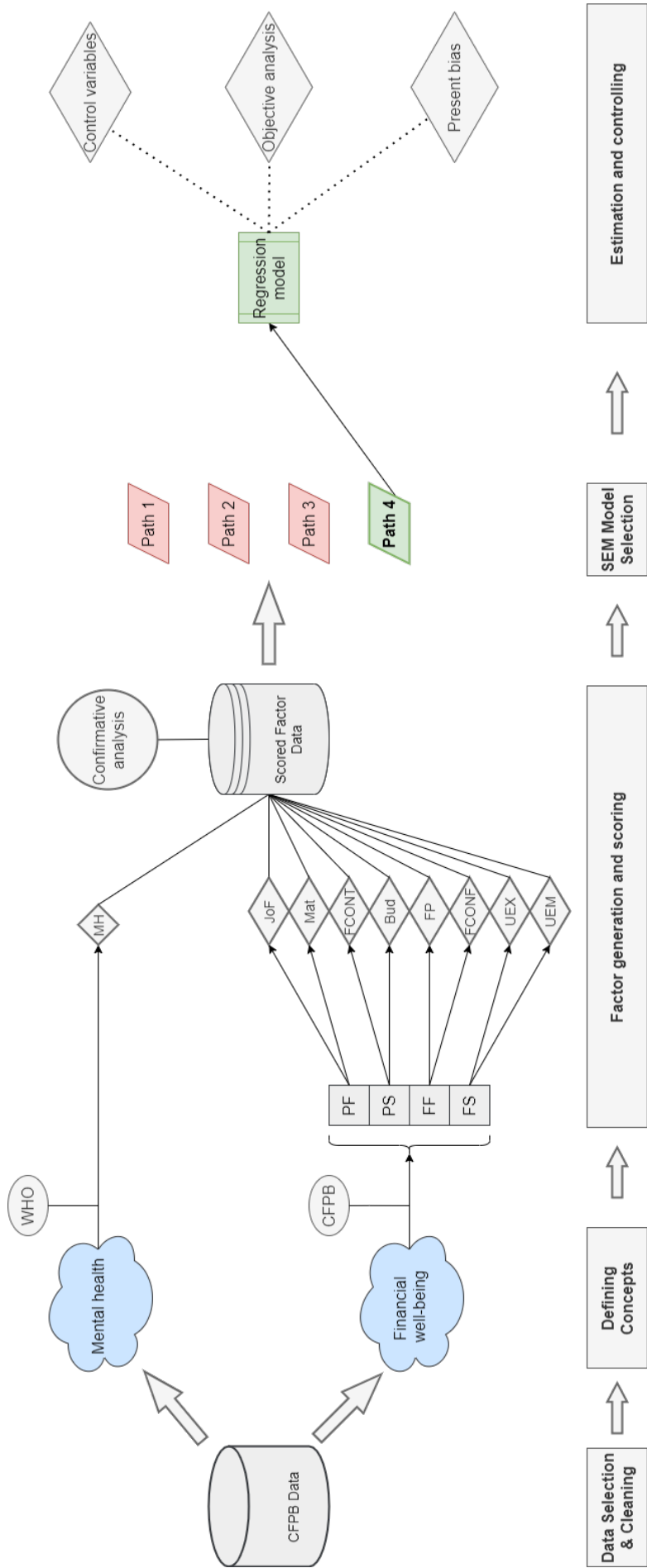
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A Appendix

Figure (A.1) Process overview



B Appendix

Linear Equations

$$\begin{aligned}
 FWB1_4 &= lvFWB1_4 f_1 \cdot f_1 + e1 \\
 FWB2_1 &= lvFWB2_1 f_1 \cdot f_1 + e2 \\
 FWB2_4 &= f_1 + e3 \\
 MATERIALISM_2 &= lvMATERIALISM_2 f_2 \cdot f_2 + e4 \\
 MATERIALISM_3 &= f_2 + e5 \\
 COLLECT &= lvCOLLECT f_3 \cdot f_3 + e6 \\
 MANAGE1_1 &= lvMANAGE1_1 f_3 \cdot f_3 + e7 \\
 MANAGE1_2 &= lvMANAGE1_2 f_3 \cdot f_3 + e8 \\
 MANAGE1_3 &= lvMANAGE1_3 f_3 \cdot f_3 + e9 \\
 ENDSMEET &= lvENDSMEET f_3 \cdot f_3 + e10 \\
 SNAP &= lvSNAP f_3 \cdot f_3 + e11 \\
 FWB1_5 &= lvFWB1_5 f_3 \cdot f_3 + e12 \\
 FWB1_6 &= lvFWB1_6 f_3 \cdot f_3 + e13 \\
 FWB2_2 &= f_3 + e14 \\
 PROPPLAN_1 &= lvPROPPLAN_1 f_4 \cdot f_4 + e15 \\
 PROPPLAN_2 &= f_4 + e16 \\
 FINGOALS &= lvFINGOALS f_5 \cdot f_5 + e17 \\
 PROPPLAN_3 &= lvPROPPLAN_3 f_5 \cdot f_5 + e18 \\
 PROPPLAN_4 &= f_5 + e19 \\
 GOALCONF &= lvGOALCONF f_6 \cdot f_6 + e20 \\
 PRODHAVE_4 &= lvPRODHAVE_4 f_6 \cdot f_6 + e21 \\
 FWB1_3 &= lvFWB1_3 f_6 \cdot f_6 + e22 \\
 FWB2_3 &= f_6 + e23 \\
 ABSORBSHOCK &= lvABSORBSHOCK f_7 \cdot f_7 + e24 \\
 SAVEHABIT &= lvSAVEHABIT f_7 \cdot f_7 + e25 \\
 PRODHAVE_6 &= lvPRODHAVE_6 f_7 \cdot f_7 + e26 \\
 FWB1_1 &= lvFWB1_1 f_7 \cdot f_7 + e27 \\
 FWB1_2 &= f_7 + e28 \\
 PRODHAVE_1 &= lvPRODHAVE_1 f_8 \cdot f_8 + e29 \\
 PRODHAVE_2 &= lvPRODHAVE_2 f_8 \cdot f_8 + e30 \\
 PRODHAVE_3 &= f_8 + e31 \\
 SWB_1 &= lvSWB_1 f_9 \cdot f_9 + e32 \\
 SWB_2 &= lvSWB_2 f_9 \cdot f_9 + e33 \\
 SWB_3 &= lvSWB_3 f_9 \cdot f_9 + e34 \\
 HEALTH &= lvHEALTH f_9 \cdot f_9 + e35 \\
 DISTRESS &= f_9 + e36 \\
 f_9 &= pf_{9,1} \cdot f_1 + pf_{9,2} \cdot f_2 + pf_{9,3} \cdot f_3 + pf_{9,4} \cdot f_4 + pf_{9,5} \cdot f_5 + pf_{9,6} \cdot f_6 + \\
 &\quad pf_{9,7} \cdot f_7 + pf_{9,8} \cdot f_8 + d_1
 \end{aligned}$$

std

$$\begin{aligned}
 f_1, f_2, \dots, f_8 &= \mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_8 \\
 d_1 &= \mathbf{V}_{d1} \\
 e1, e2, \dots, e36 &= \mathbf{V}_{e1}, \mathbf{V}_{e2}, \dots, \mathbf{V}_{e36}
 \end{aligned}$$

covariance

$$\begin{aligned} f_1 \cdot f_2 &= \mathbf{S}_{1,2} \\ f_1 \cdot f_3 &= \mathbf{S}_{1,3} \\ f_1 \cdot f_4 &= \mathbf{S}_{1,4} \\ f_1 \cdot f_5 &= \mathbf{S}_{1,5} \\ f_1 \cdot f_6 &= \mathbf{S}_{1,6} \\ f_1 \cdot f_7 &= \mathbf{S}_{1,7} \\ f_1 \cdot f_8 &= \mathbf{S}_{1,8} \\ f_2 \cdot f_3 &= \mathbf{S}_{2,3} \\ f_2 \cdot f_4 &= \mathbf{S}_{2,4} \\ f_2 \cdot f_5 &= \mathbf{S}_{2,5} \\ f_2 \cdot f_6 &= \mathbf{S}_{2,6} \\ f_2 \cdot f_7 &= \mathbf{S}_{2,7} \\ f_2 \cdot f_8 &= \mathbf{S}_{2,8} \\ f_3 \cdot f_4 &= \mathbf{S}_{3,4} \\ f_3 \cdot f_5 &= \mathbf{S}_{3,5} \\ f_3 \cdot f_6 &= \mathbf{S}_{3,6} \\ f_3 \cdot f_7 &= \mathbf{S}_{3,7} \\ f_3 \cdot f_8 &= \mathbf{S}_{3,8} \\ f_4 \cdot f_5 &= \mathbf{S}_{4,5} \\ f_4 \cdot f_6 &= \mathbf{S}_{4,6} \\ f_4 \cdot f_7 &= \mathbf{S}_{4,7} \\ f_4 \cdot f_8 &= \mathbf{S}_{4,8} \\ f_5 \cdot f_6 &= \mathbf{S}_{5,6} \\ f_5 \cdot f_7 &= \mathbf{S}_{5,7} \\ f_5 \cdot f_8 &= \mathbf{S}_{5,8} \\ f_6 \cdot f_7 &= \mathbf{S}_{6,7} \\ f_6 \cdot f_8 &= \mathbf{S}_{6,8} \\ f_7 \cdot f_8 &= \mathbf{S}_{7,8} \end{aligned}$$

Table (B.1) Factor decomposition

PF		PS		FF		FS		MH
JoF	Mat	FCONT	Bud	FPL	FCONF	UEX	UEM	MH
FWB1_4	MATERIALISM_2	COLLECT	MANAGE1_2	FINGOALS	FWB1_3	ABSORBSHOCK	PRODHAVE_1	SWB_1
FWB2_1	MATERIALISM_3	ENDSMEET	PROPPLAN_1	PROPPLAN_3	FWB2_3	FWB1_1	PRODHAVE_2	SWB_2
FWB2_4		FWB1_5	PROPPLAN_2	PROPPLAN_4	GOALCONF	FWB1_2	PRODHAVE_3	SWB_3
		FWB1_6		PRODHAVE_4		PRODHAVE_6		DISTRESS
		FWB2_2				SAVEHABIT		GOALCONF
		FWB2_2						HEALTH
		MANAGE1_1						
		MANAGE1_2						
		MANAGE1_3						
		SNAP						

Table (B.2) Variable description

Variable	Label
ABSORBSHOCK	Confidence in ability to raise \$2,000 in 30 days
BENEFITS_5	Work/Life Benefits
CHANGEABLE	Belief that ability to manage money is NOT changeable
COLLECT	Contacted by debt collector in past 12 months
DISTRESS	Lot of stress in respondent's life
EARNERS	Household members currently contributing to paying household bills
ENDSMEET	Difficulty of covering monthly expenses and bills
FINGOALS	Do you have a current or recent financial goal?
FRUGALITY	If I can re-use an item I already have, there's no sense in buying something new
FWB1_1	I could handle a major unexpected expense
FWB1_2	I am securing my financial future
FWB1_3	Because of my money situation...I will never have the things I want in life
FWB1_4	I can enjoy life because of the way I'm managing my money
FWB1_5	I am just getting by financially
FWB1_6	I am concerned that the money I have or will save won't last
FWB2_1	Giving a gift...would put a strain on my finances for the month
FWB2_2	I have money left over at the end of the month
FWB2_3	I am behind with my finances
FWB2_4	My finances control my life
GOALCONF	Confidence in own ability to achieve financial goals
HEALTH	In general, would you say your health is . . .
MANAGE1_1	Paid all your bills on time
MANAGE1_2	Stayed within your budget or spending plan
MANAGE1_3	Paid off credit card balance in full each month
MATERIALISM_2	The things I own say a lot about how well I'm doing in life
MATERIALISM_3	I like to own things that impress people
MEMLOSS	Experienced memory loss or confusion in past 12 months
PROPPLAN_1	I consult my budget to see how much money I have left
PROPPLAN_2	I actively consider the steps I need to take to stick to my budget
PRODHAVE_1	Checking or Savings Account at a bank or credit union
PRODHAVE_2	Life Insurance
PRODHAVE_3	Health Insurance
PRODHAVE_4	Retirement Account (such as a 401k or IRA)
PRODHAVE_5	Pension
PRODHAVE_6	Non-Retirement Investments (such as stocks, bonds or mutual funds)
PROPPLAN_3	I set financial goals for what I want to achieve with my money
PROPPLAN_4	I prepare a clear plan of action w/ detailed steps to achieve my financial goals
SAVEHABIT	Putting money into savings is a habit for me
SELFCONTROL_1	I often act without thinking through all the alternatives
SELFCONTROL_2	I am good at resisting temptation
SELFCONTROL_3	I am able to work diligently toward long-term goals
SHOCKS_11	Provided unexpected financial support to a family member or friend
SNAP	Any household member received SNAP benefits
SWB_1	I am satisfied with my life
SWB_2	I am optimistic about my future
SWB_3	If I work hard today, I will be more successful in the future
VOLATILITY	Household income volatility